# MASTER OF COMPUTER APPLICATIONS

## PRACTICAL RECORD WORK
## ON
## 20MCA136- NETWORKING & SYSTEM ADMINISTRATION LAB

**SUBMITTED BY**

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**DEPARTMENT OF COMPUTER APPLICATIONS**

**COLLEGE OF ENGINEERING VATAKARA**

**(CAPE - GOVT. OF KERALA)**

**APRIL 2026**

**DEPARTMENT OF COMPUTER APPLICATIONS**

**COLLEGE OF ENGINEERING VATAKARA**

**(CAPE -GOVT.OF KERALA)**

# CERTIFICATE

Certified that this is the Bonafide record work on the practical course **20MCA136 NETWORKING & SYSTEM ADMINISTRATION LAB** done and prepared by Mr. **MUHAMMED NAJI A C (Reg.No VDA25MCA-2027), 2nd Semester MCA (2025 -27 Batch) student of Department of Computer Applications at College of** Engineering Vadakara, in the partial fulfilment of the award of MCA Degree of APJ Abdul Kalam Technological University (KTU).

Date:

Faculty-in-Charge Head of the Department

(Office Seal)

Internal Examiners: External Examiners:

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# EXPERIMENT 1

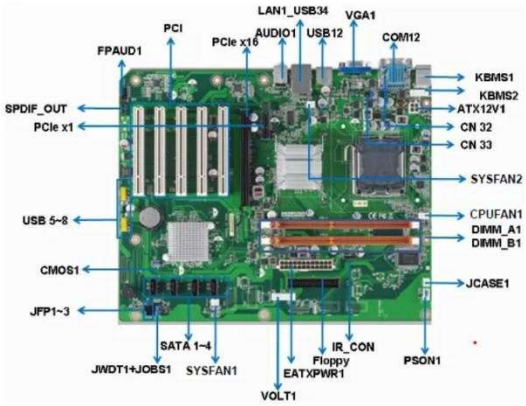
**Aim:** To identify the major components of a computer system such as motherboard, ram modules, daughter cards, SMPS, busloads, internal storage devices and interfacing ports. Specificationof desktop and server class computers. Installation of common operating system for desktop and server use.

## COMPUTER HARDWARE

Computer hardware refers to the physical parts or components of a computer such as the monitor, mouse, keyboard, computer data storage, hard drive disk (HDD), system unit (graphic cards, sound cards, memory, motherboard and chips), etc. all of which are physical objects that can be touched.

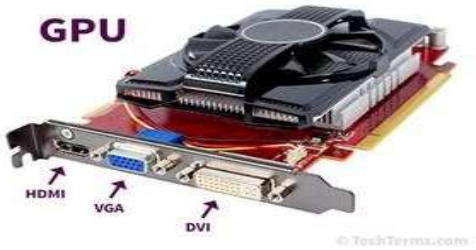
## MOTHERBOARD

The motherboard serves as a single platform to connect all of the parts of a computer together. It connects the CPU, memory, hard drives, optical disk, video card, sound card, and other ports and expansion cards directly or via cables. It can be considered as the backbone of a computer. Motherboard contains ports to connect all of the internal components. It holds together many of the crucial components of a computer, including central processing unit, memory and connectors for input and output devices.




## GPU(GRAPHIC PROCESSING UNITS)

GPUs also known as graphic cards or video cards or graphics cards. In order to display pictures, videos, audios, and 2D or 3D animations, each device uses a GPU.A GPU performs fast calculations of arithmetic and frees up the CPU to do different things




**CMOS (COMPLEMENTARY METAL OXIDE SEMICONDUCTOR)**

CMOS is a combination of NMOS and PMOS transistors that operates under the applied electrical field. The structure of CMOS was initially developed for high density and low power logic gates.

**HDMI (HIGH-DEFINITION MULTIMEDIA INTERFACE)**

It is an all-digital audio-video interface which transmits signals in uncompressed format,
Eg: uncompressed video data and compressed or uncompressed digital data from a HDMI-compliant source device to a computer monitor, video projector, digital television…etc.

**SMPS**

SMPS is an electronic power supply system that makes use of a switching regulator to transfer electrical power effectively. It is a PSU (POWER SUPPLY UNIT) and is usually used in computers to change the voltage to the appropriate range for the computer.

**DGA (DOMAIN GENERATION ALGORITHM)**

It is a technique used by cyber attackers to generate new domain names and IP addresses for malware's command and control servers.

## COOLING FAN

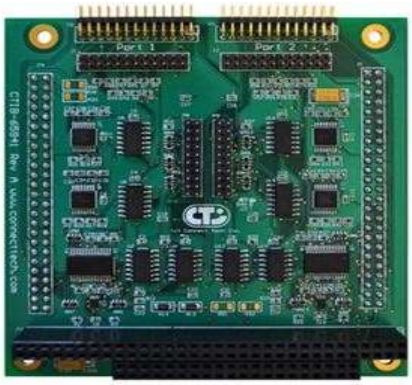
A computer is any fan inside, or attached to, a computer case used for active cooling. Fans are used to draw cooler air into the case from the outside. Expel warm air from inside and move air across a heat sink to cool a particular component.

## RAM MODULES

In computing, a memory module or RAM (random access memory) stick is a printed circuit board on which memory integrated circuit are mounted. Memory modules permit easy installation and replacement in electronic systems, especially computers such as personal computers, work stations, and servers. The first memory modules were proprietary designs that were specific to a model of computer from a specific manufacturer.

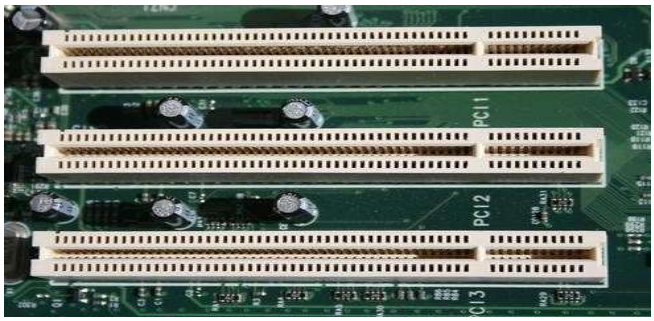
## DAUGHTER CARDS

A daughterboard (or daughter board, daughter card, or daughtercard) is a circuit board that plugs into and extends the circuitry of another circuit board. The other circuit board may be the computer's main board (its motherboard) or it may be another board or card that is already in the computer, often a sound card. The term is commonly used by manufacturers of wavetable daughter boards that attach to existing sound cards.



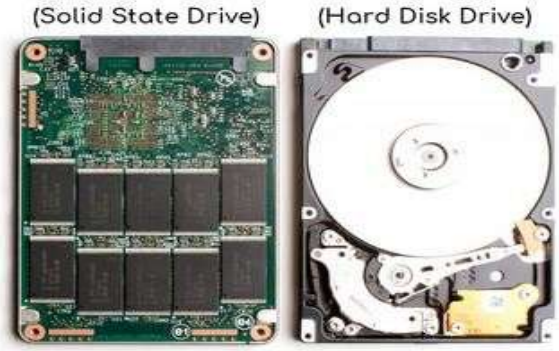
**BUS SLOT**

Alternatively known as a bus slot or expansion port, an expansion slot is a connection or port inside a computer on the motherboard or riser card. It provides an installation point for a hardware expansion card to be connected. For example, if you wanted to install a new video card in the computer, you'd purchase a video expansion card and install that card into the compatible expansion slot.




**STORAGE DEVICES**

A storage device is a kind of hardware, which is also known as storage, storage medium, digital storage, or storage media that has the ability to store information either temporarily or permanently. It is used to hold, port, and extract data files.




**1)SSD (SOLID STATE DRIVE)**

SSD is non-volatile storage device, it stores the data on flash memory chips and maintains the data in a permanent state, even when the power is off. As compared to electromechanical drives,

SSDs have lower latency and access quickly. These storage devices store the data in the semiconductor cells.
Following are several types of SSDs:

**SATA SSD**
SATA is the acronym for 'serial advanced technology attachment'. Speaking of SSD type SSD is the most popular today.

**M.2 SATA SSD:**
Its newer. and its format is lighter and smaller than the SATA SSD

**mSATA SSD:**
Its mini version of SATA. It has a smaller form factor mainly used in ultra-compact computers, laptops, mobile devices with an mSATA slot, in which the installation of an extended size SATA SSD is impossible.

**2)HDD (HARD DISK DRIVE):**
HDD is an electro mechanical storage device, which is an abbreviation of hard disk drive. It uses magnetic storage for storing and retrieving the digital data. It is a non-volatile storage device. Hard disk is installed internally in our computer systems, which is connecter directly to the disk controllers of the motherboard.

HDD means data is retained when our computer system is shutdown.

## INPUT DEVICES/UNITS

An input device is essentially a piece of instrument or hardware that allows users to provide data, information, or control instructions to a computer used for interaction and control. Data is entered into a computer in a raw format, which is converted into computer understandable language by input devices and processed by a central processing unit (CPU) to produce output. Some common types of input devices are:

**1)KEYBOARD:**

The keyboard is one of the primary input devices, which helps in entering data and commands in a computer. A normal keyboard usually has a variety of keys, such as alphabetic character keys, function keys, number keys, arrow keys, and control keys. The keyboard can be connected to a computer using USB or BLUETOOTH.

**2)MOUSE**

Mouse is the most common and very popular pointing device that helps interact with a common through a process called 'point and click'. This is mainly used to move a cursor on the computer's screen and click on the corresponding object using buttons (usually left, right, and middle key roller buttons).

**3)SCANNER**

A scanner is an input device that works like a photocopy machine. It is used when information is available on paper and needs to be transferred to the computer's hard disk for further manipulation.

## PROCESSING UNIT

The part of a computer that performs logical and arithmetical operation on the data as specified in the instructions.

**1)CPU (CENTRAL PROCESSING UNIT)**

A central processing unit is also called a processor, central processor, or microprocessor. It carries out all the important functions of a computer. It stores all important programs like operating systems and application software. It is often referred to as the brain of the computer.

**2)RAM (RANDOM ACCESS MEMORY)**

Is a hardware device generally located on the motherboard of a computer of the CPU. It allows CPU store data, program, and program result when you switch on the computer.it is the read and write memory of a computer, which means the information can be written to it as well as read from it.

## OUTPUT DEVICES/UNITS

The output device displays the result of the processing of raw data that is entered in the computer through an input device. There are a number of output devices that displays output in different ways such as text, images, hard copies, and audio or video. Some popular output devices are:

**1)MONITOR:**

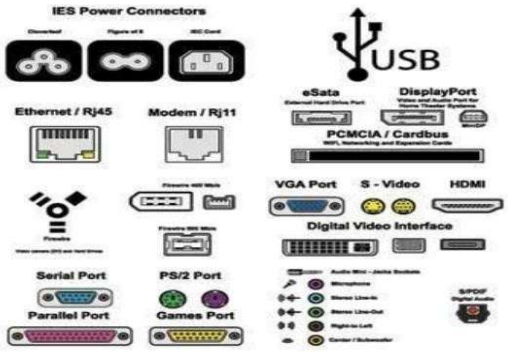
A monitor is a piece of computer hardware that accepts data from a computer and displays it on the system screen through the computers video card. Monitors have the ability to display information at

much higher resolution. Additionally, these are much like televisions and also known as video screen, display, video display terminal, or video display unit.

**2)SPEAKERS**:
The most common output devices, speakers accept sound data from a computer and play the sounds for users to hear.

**3)PROJECTOR:**
Projector is an output device that accepts data from a computer and projects that data or information as a picture onto a wall or screen or any large surface.

**INTERFACING PORTS:**
A port is a physical docking point using which an external device can be connected to the computer. It can also be programmatic docking point through which information flows from a program to the computer or over the Internet.




## DESKTOP:

A desktop computer is a personal computer designed for regular use at a single location on or near a desk due to its size and power requirements. The most common configuration has a case that houses the power supply, motherboard (a printed circuit board with a microprocessor as the central processing unit, memory, bus, certain peripherals and other electronic components), disk storage (usually one or more hard disk drives, solid state drives, optical disc drives, and in early models a floppy disk drive); a keyboard and mouse for input; and a computer monitor, speakers, and, often, a printer for output. The case may be oriented horizontally or vertically and placed either underneath, besides, or on top of a desk.

## SERVER OPERATING SYSTEM

A server operating system is a type of operating system that is designed to be installed and used on a server computer. It is advanced version of operating system, having features and capabilities required within a client-server architecture or similar enterprise computing environment.
Example: Windows Operating System, Linux Operating System

**DATA SERVER:**
A data server is a software program/platform used to provide database service like storing, processing, and securing data. Mainly three types:

**FILE SERVERS:**
File servers store and distribute files. Multiple clients or users may share files stored on a server. In addition, centrally storing files offers easier backup or fault tolerance solutions than attempting to provide security and integrity for files on every device in an organization. File server hardware can be designed to maximize read and write speeds to improve performance.

**MAIL SERVERS**:
Mail servers are a very common type of application server. Mail servers receive emails sent to a user and store them until requested by a client on behalf of said user. Having an email server allows for a single machine to be properly configured and attached to the network at all times. It is then ready to send and receive messages rather than requiring every client machine to have its own email subsystem continuously running.

**WEB SERVERS:**
One of the most abundant types of servers in today's market is a web server. A web server is a special kind of application server that hosts programs and data requested by users across the Internet or an intranet. Web servers respond to requests from browsers running on client computers for web pages, or other web-based services. Common web servers include Apache web servers, Microsoft Internet Information Services (IIS) servers and Nginx servers. Mainly three forms are:

**BLADE SERVERS:**
The original computer server hardware was large and stored in racks that could hold hundreds of pounds. Over time, however, faster means of connecting hardware resulted in parts of the server being extracted from a single self-contained device. By removing hard drives, eliminating internal cooling, and the ongoing miniaturization of computing parts, servers were eventually reduced to a single thin server known as a blade server. While still stored in racks in server rooms, blade servers are smaller and can be replaced more easily.

**RACK SERVERS**:

A rack server, also known as a rack mount server, rack-mounted server or rack mount computer, is a computer designed to be situated in a rectangular structure called a server rack. The advantages of a server rack include better space conservation for rack servers, increased scalability, maximized air flow when coupled with a cooling system and ease of regular computer maintenance and diagnostics, given that their design allows technicians and operators to easily slide rack servers in and out of them.

**TOWER SERVERS:**

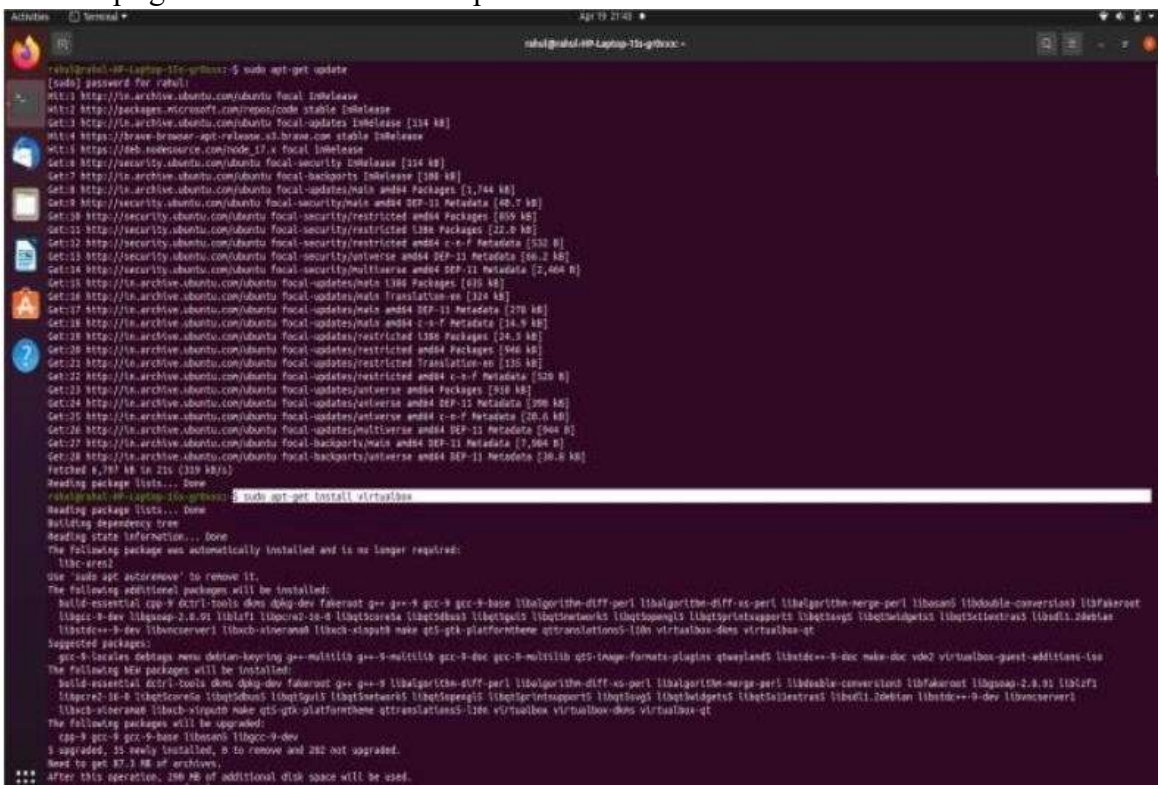
A tower server is a computer intended for use as a server and built in an upright cabinet that stands alone. The cabinet, called a tower, is similar in size and shape to the cabinet for a tower-style personal computer. This is in contrast to rack servers or blade servers, which are designed to be rack-mounted. Advantages of tower servers include: Easier cooling, because the overall component density is fairly low. Scalability, because an unlimited number of servers can be added to an existing network.

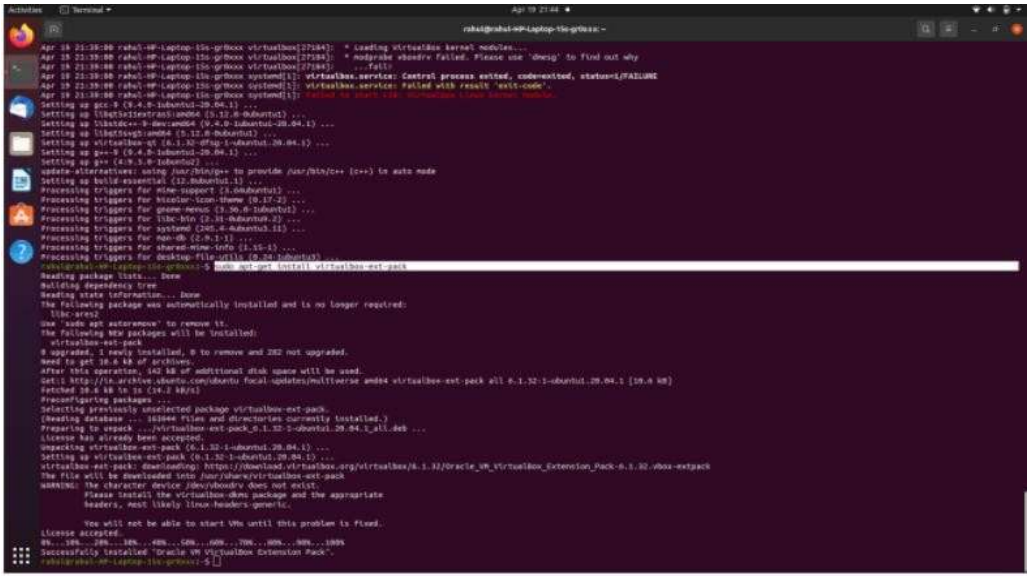
# EXPERIMENT 2

## INSTALL UBUNTU ON VIRTUALBOX

Oracle VM VirtualBox is free and open-source and being developed by oracle corporation. Oracle VirtualBox is a cross-platform virtualization application. It installs on your existing Intel or AMD-based computers, whether they are running Windows, Mac, Linux or Solaris operating systems. VirtualBox can create and run a "guest" operating system (virtual machine) in a window of the host operating system. The virtual machine provides a self-contained environment in which to experiment with new software without risking damaging changes to the host operating system VirtualBox Installation:

- sudo apt-get install virtualbox
- sudo apt-get install virtualbox-ext-pack





Create virtual machine by just clicking on this new Click -> new

we can install ubuntu so type ubuntu And choose the type

Click Next

Click next

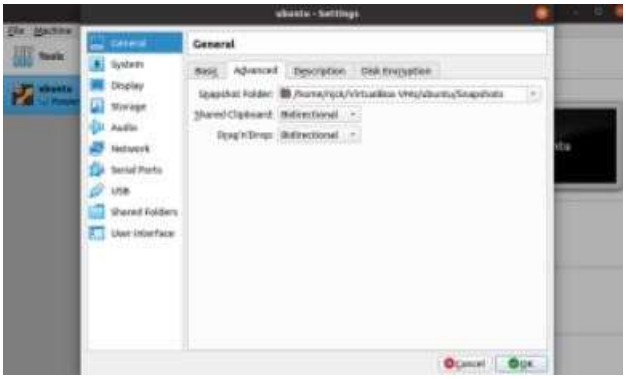
Select the Virtual hard disk

Click-> Next

Select Dynamically allocated memory. Click->Next

Select size of the virtualbox

Click->create



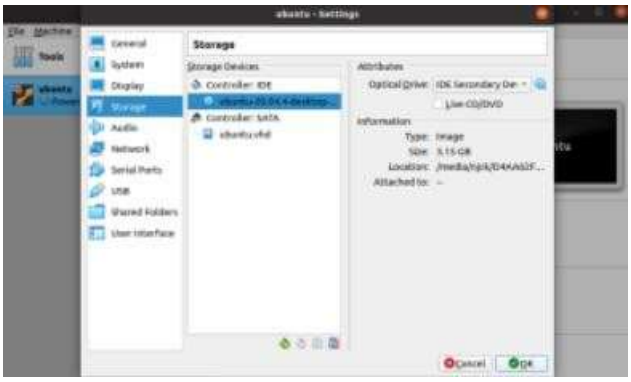

Settings - > General-> Advanced

Set Shared clipboard and drag ’n’ Drop as Bidirectional.

Click -> ok

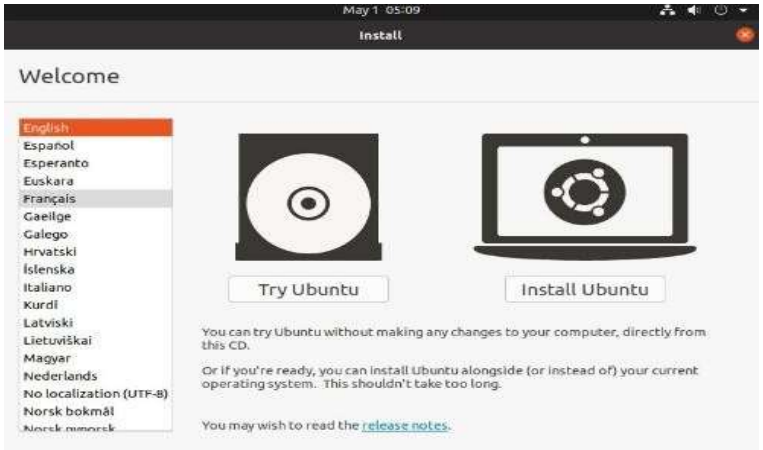
Download Ubuntu from

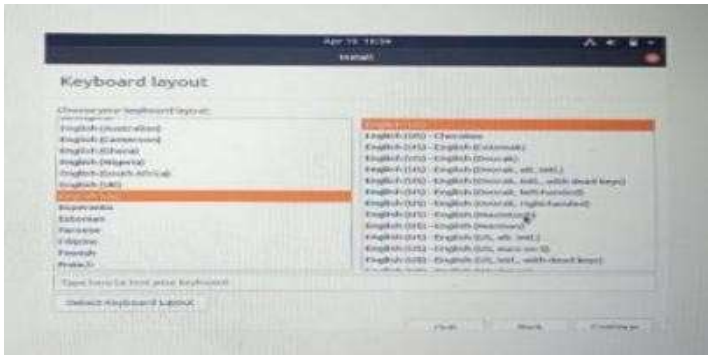
https://ubuntu.com/download/desktop/ this site.



Settings->Storage->Attributes->optical Drive select downloaded ubuntu iso file. Click -> ok

Click-> Start and install ubuntu

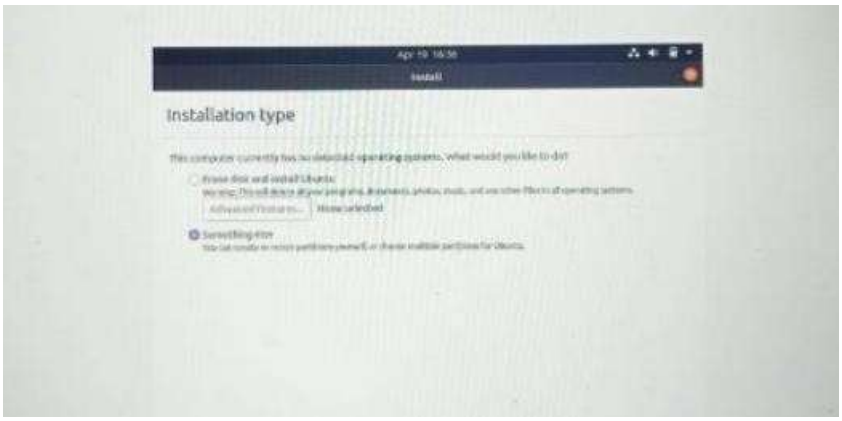



Click - > Continue

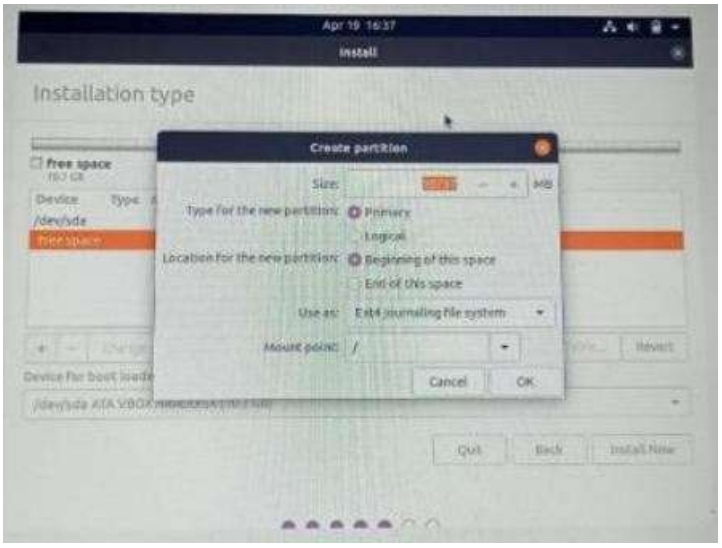
Select Normal installation and Download updates while installing ubuntu.

Click->continue

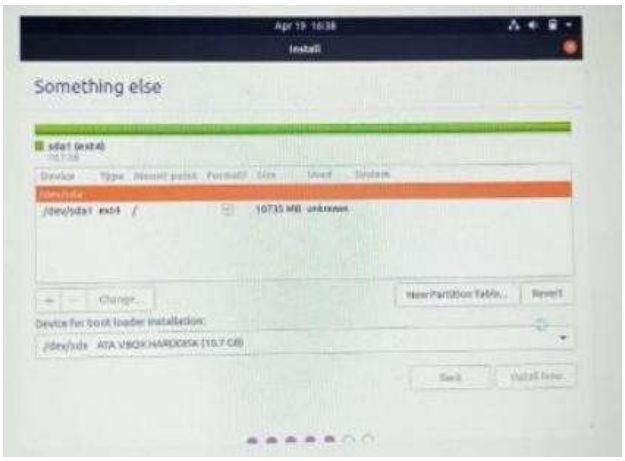



Select installation type something else.

Click->continue

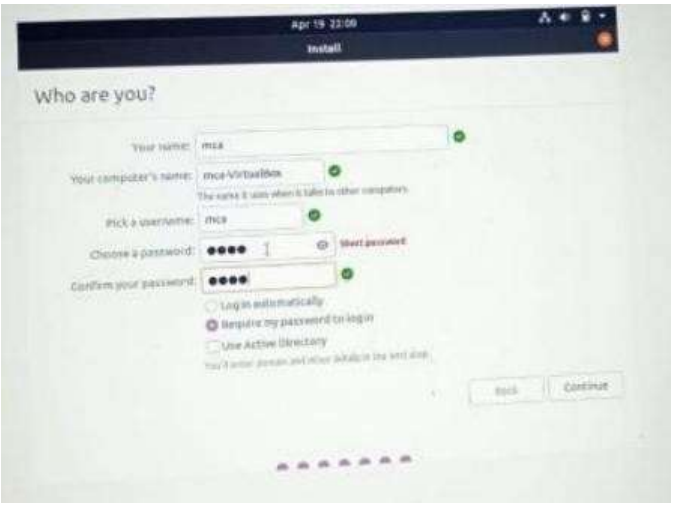


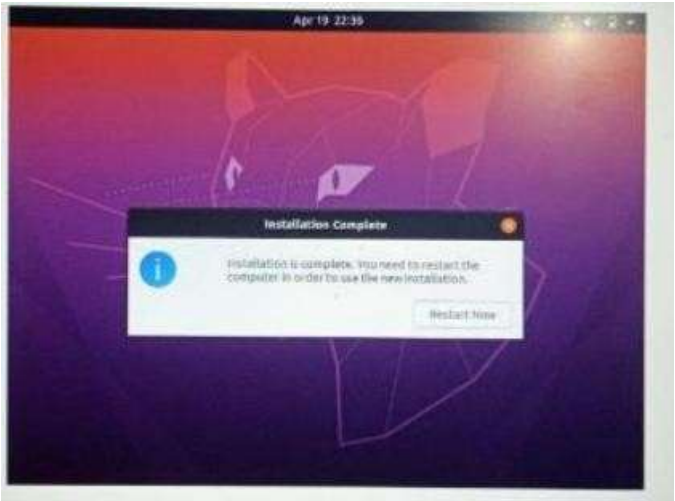

Select disk then create partition and set mount point as '/' click -> ok



click -> install now

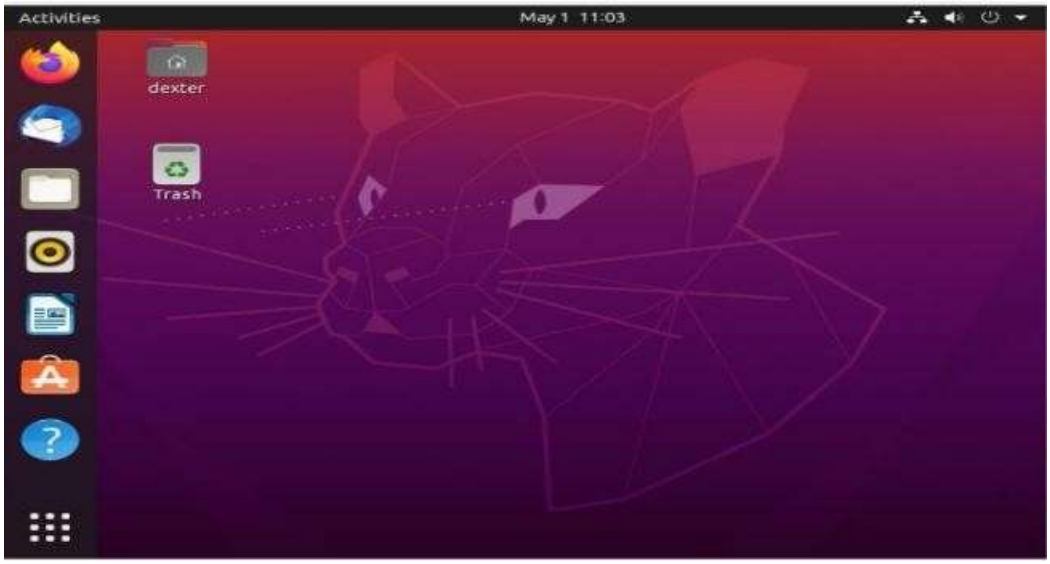
click -> continue




Set Your name and Password
click -> continue

Click reinstall now



# SERVER OPERATING SYSTEM

## [CentOS]

Linux is an open-source operating system currently powering most of the Internet. There are hundreds of different versions of Linux. For web servers, the two most popular versions are Ubuntu and CentOS. Both are open-source and free community supported operating systems.

**CentOS (Community Enterprise Operating System)** is a free, open-source Linux distribution. This community-built software project, created by Red Hat Enterprise Linux, was initially released in 2004. The first release was forked from RHEL version 2.1AS using its source code. It provided an enterprise-level computing platform freely available for anyone to use. A new version of CentOS has been released after each new RHEL distribution with a one-to-two-month lag as a downstream community project.

**RESULT:**
We identify the major component of a computer system. That is motherboard, input devices, output devices and storage devices etc.…

**Install virtual box;** virtual box allows users and administrators to easily run multiple guest operating systems on a single host. We identify that a desktop computer system typically runs a user-friendly operating system and desktop applications to facilitate desktop-oriented tasks. In contrast, a server manages all network resources.

# EXPERIMENT 3

**Aim:** Study of a terminal based text editor such as Vim or Emacs. (By the end of the course, students are expected to acquire following skills in using the editor: cursor operations, manipulate text, search for patterns, global search and replace) Basic Linux commands, familiarity with following commands/operations expected.

1. man
2. ls, echo, read
3. more, less, cat,
4. cd, mkdir, pwd, find
5. mv, cp, rm, tar
6. wc, cut, paste
7. head, tail, grep, expr
8. chmod, chown
9. Redirections & Piping
10. useradd, usermod, userdel, passwd
11. df, top, ps
12. ssh, scp, ssh-keygen, ssh-copy-id

## TEXT EDITOR

A text editor is a type of computer program that edits plain text Text editors are provided with operating systems and software development packages, and can be used to change files such as configuration files, documentation files and programming language source code.

Unix text editors are:

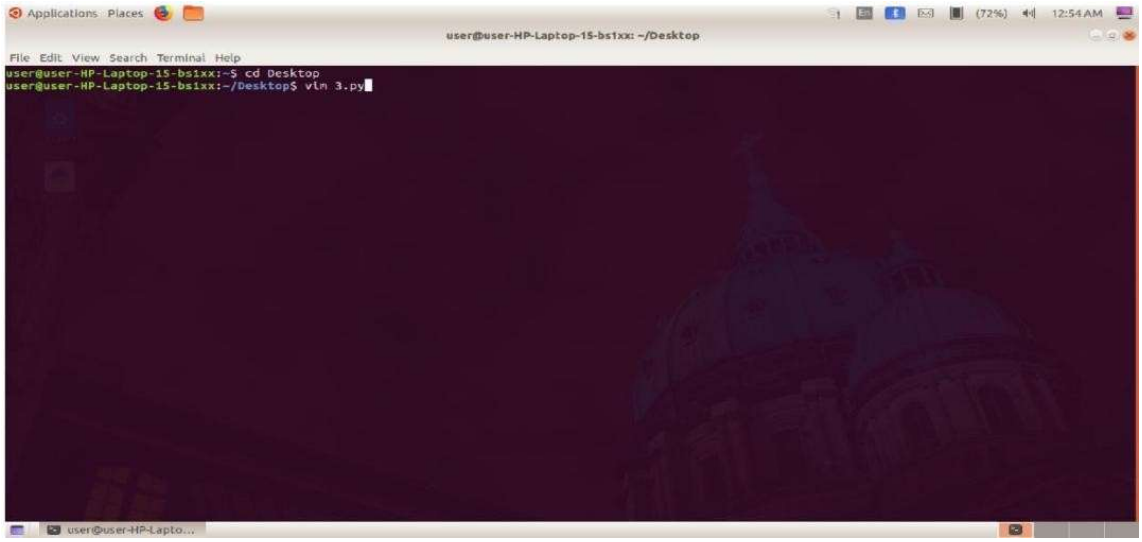
- VIM
- EMACS
- NANO
- PICO

**VIM**

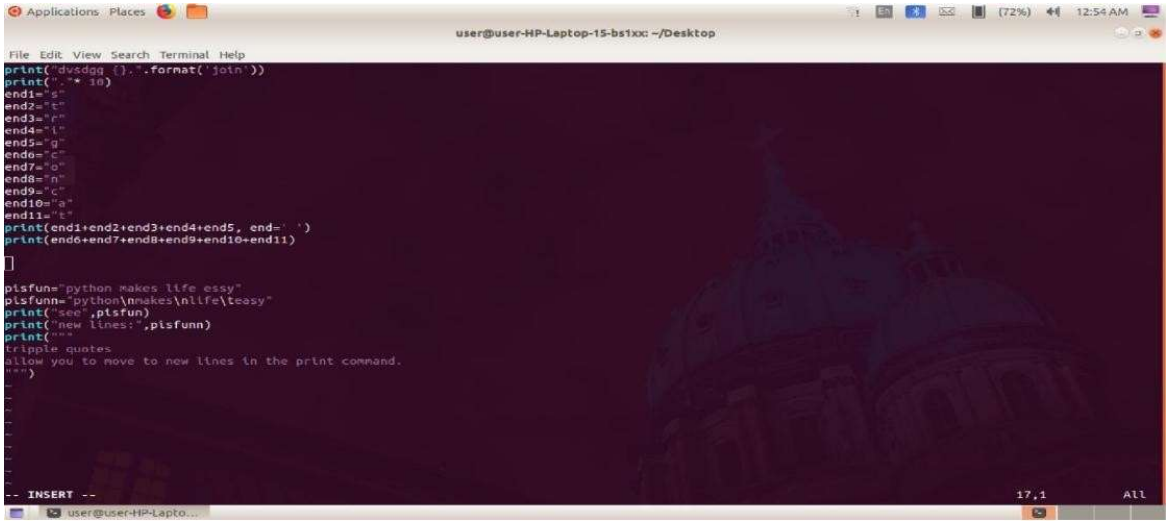
A text editor, written by Bill Joy in 1976. Short for Visual Interface. It enables fast, simple, and effective text editing mostly based on simple key bindings. It provides fast and convenient moving around files and between files. One must learn a good number of commands to be proficient in vi. Vim has two modes:

- Command
- Input

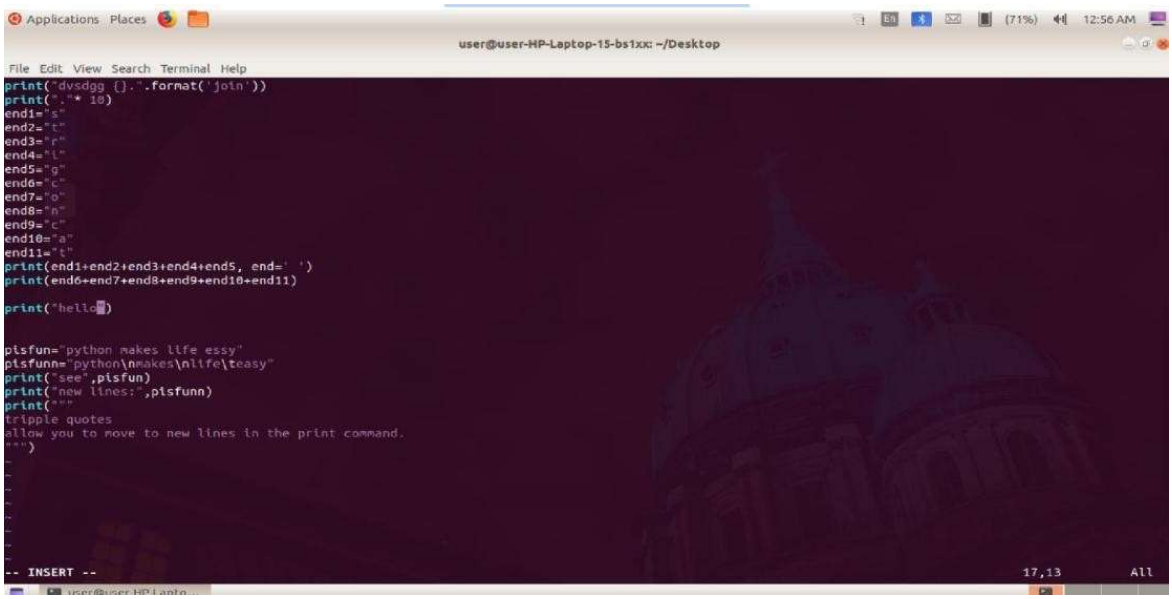
**Command mode:** Allows you to do global operations likesaving the file, searching for a string and replacing it. Used to enter commands. It allows Text manipulation **Insert mode/Input mode:** The mode that lets you edit and enter the text. Hitting 'I' will get you to insert mode from command mode.



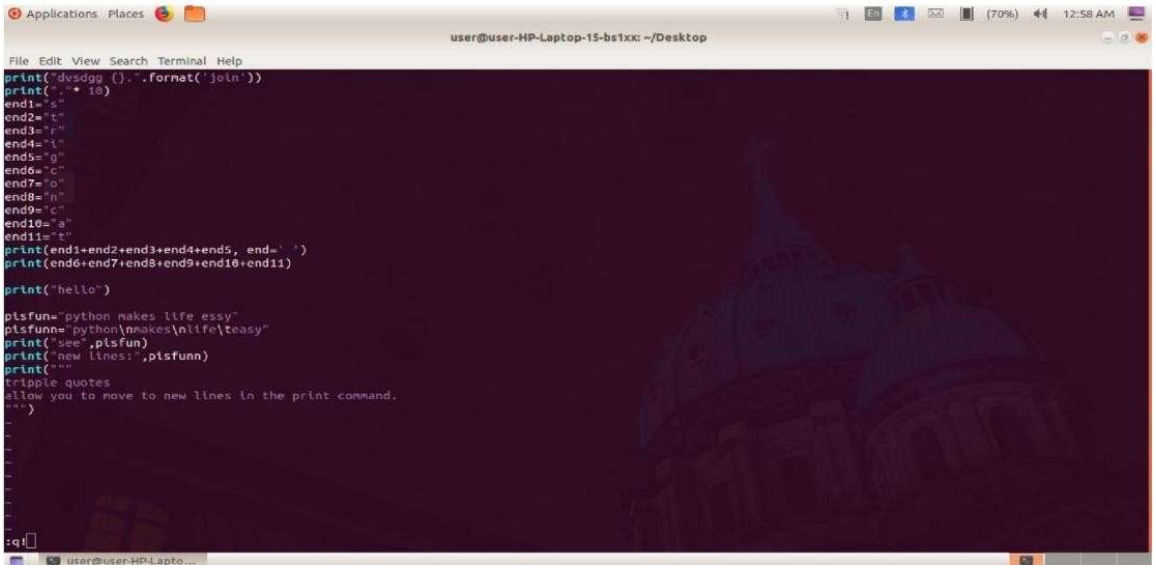

**The file 3.py opened and insert mode activated**



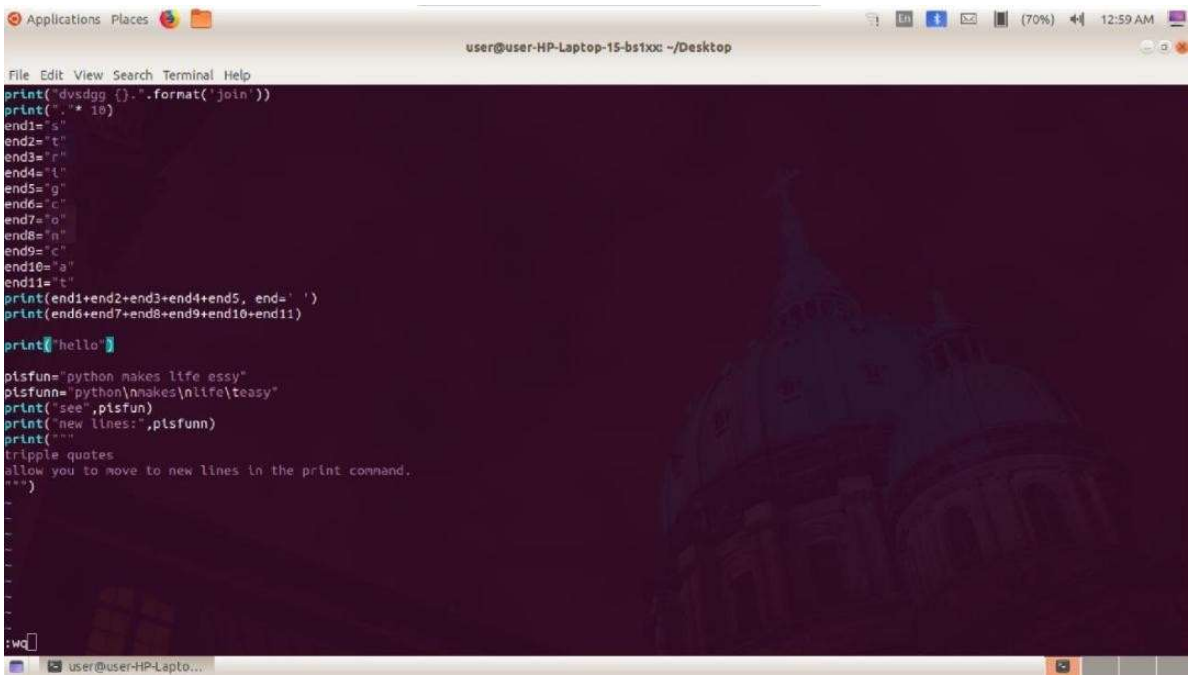

**The file is gotten edited**



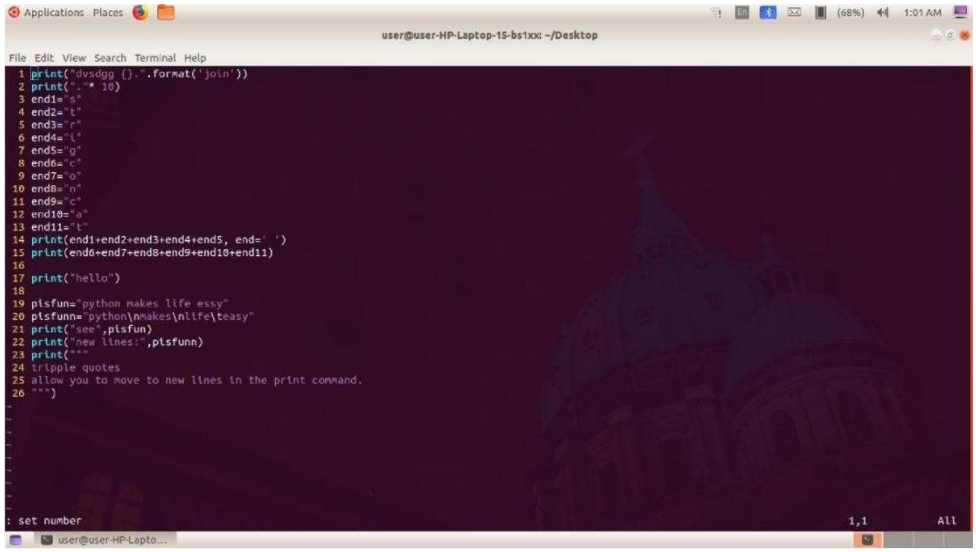

**ESCAPE : q! [ Exit without save ]**




**ESCAPE : wq [ Save and Exit ]**

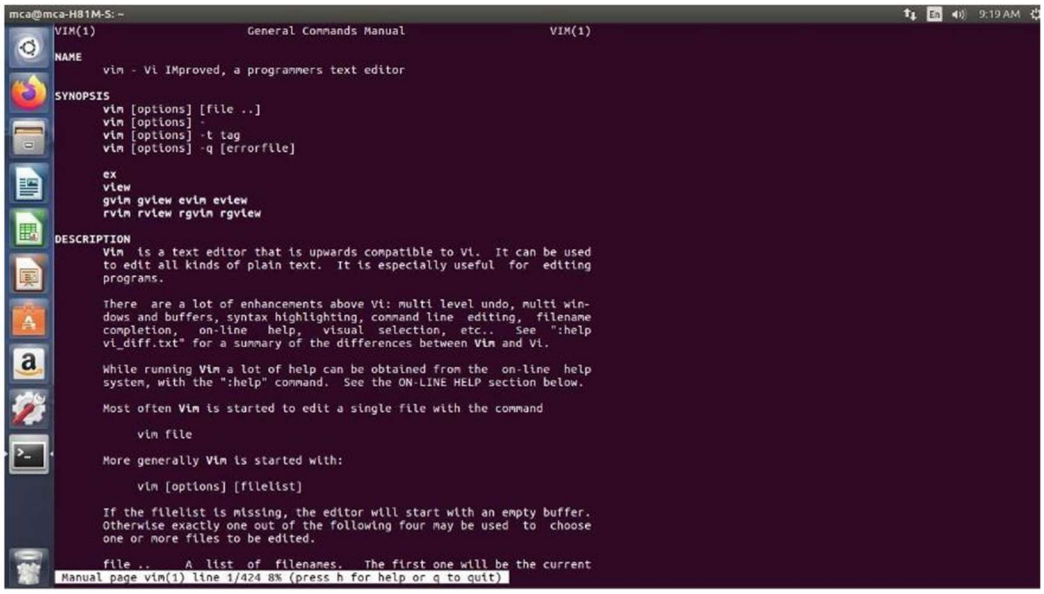



**ESCAPE : set number [Set number to the lines]**



## BASIC LINUX COMMANDS

**1)man:** Show the manual for a given command.

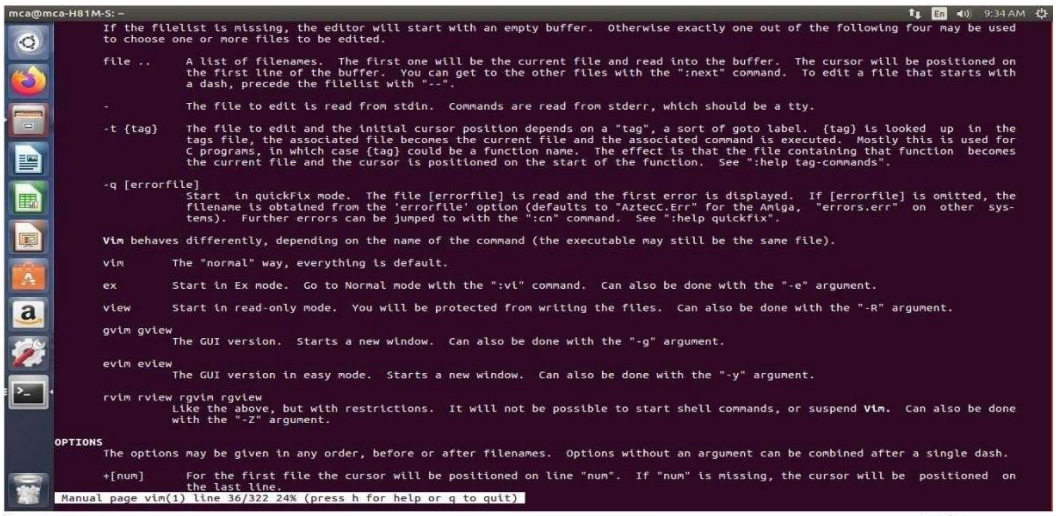


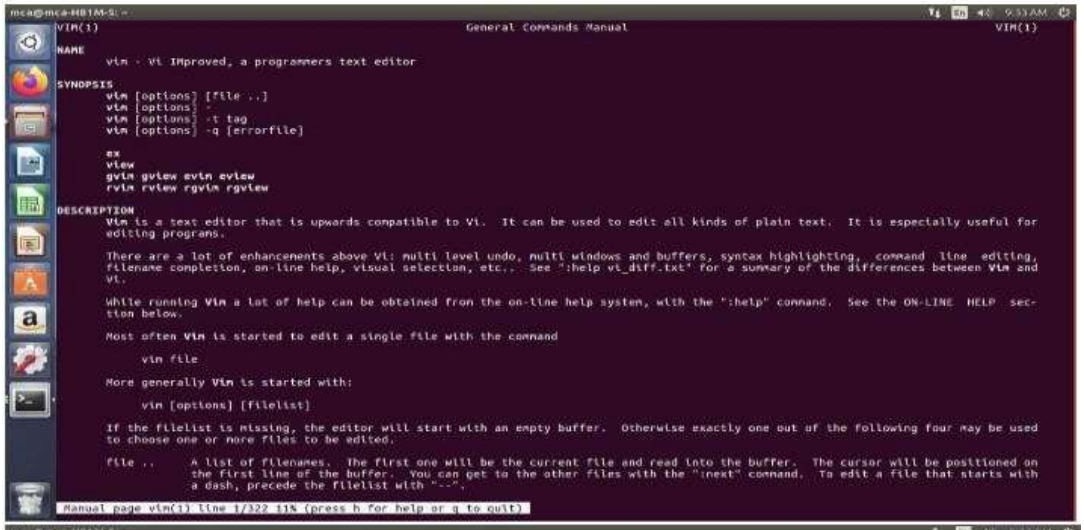

**2)ls:** List the directory(folder) system.
ls -a: Will show the hidden file. ls -al: Will list the file and directory with detailed information like the permission size,owner…etc
For example, enter ls /home/username/Documents to view the content of Documents.

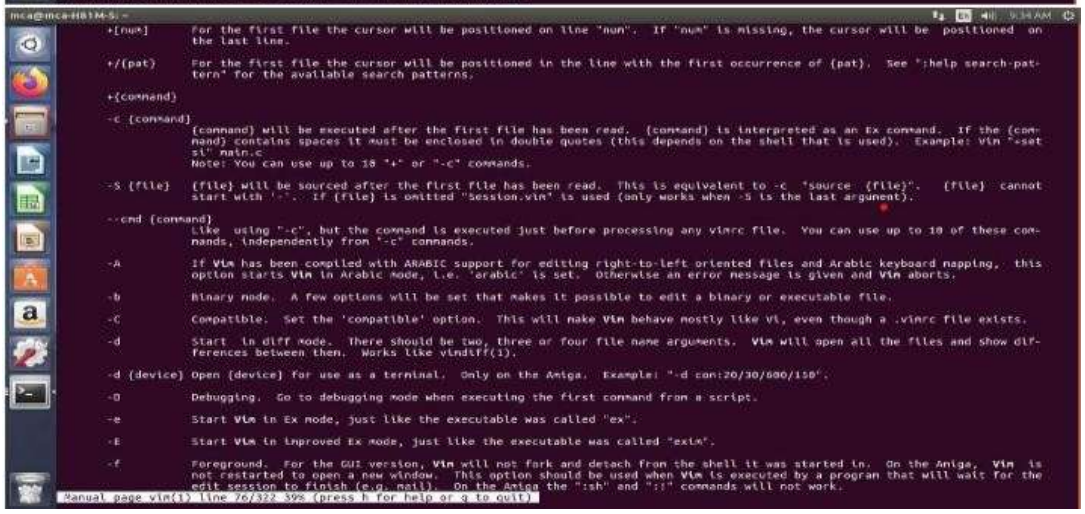
**3)echo:** It is built in linux feature that print out arguments as the standard output.

**4)read:** It is used to read the contents of a line into a variable.

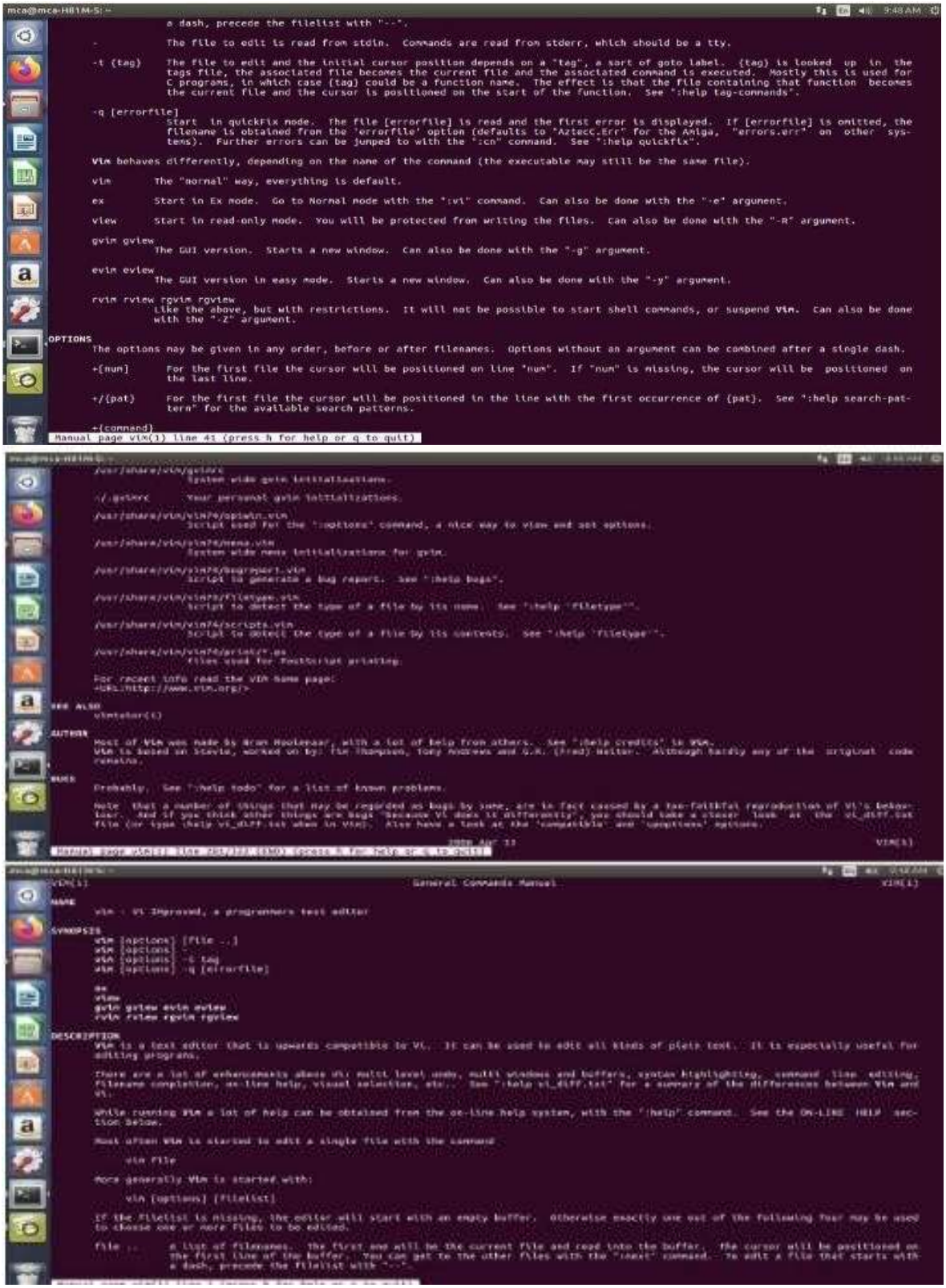
**5)more**: It is used to view the text files in the command prompt,displaying one screen at a time in case the file is large.



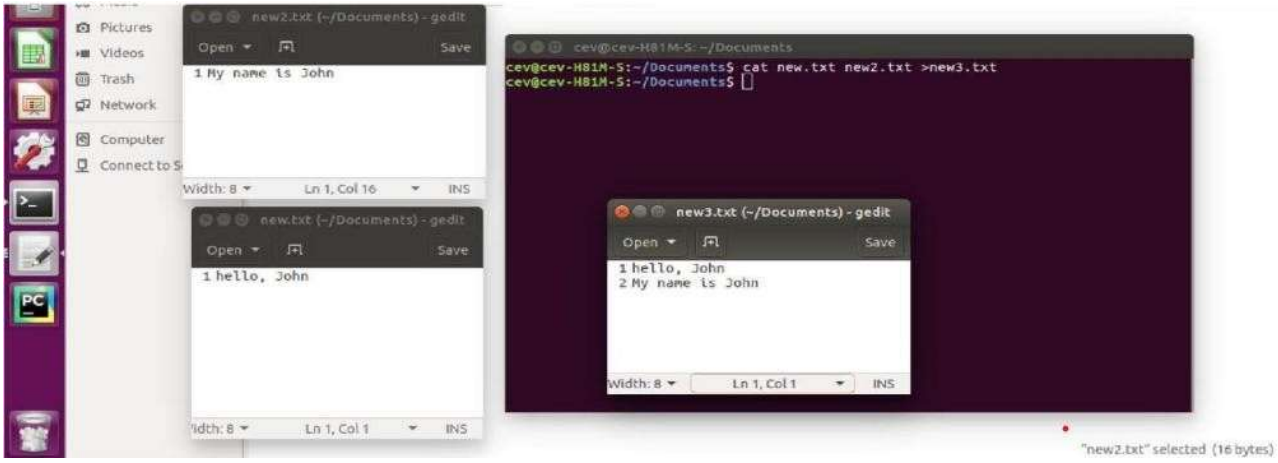




**6)less:** Less command is a linux utility that can be used to read the contents of a text file one page(one screen) at a time.



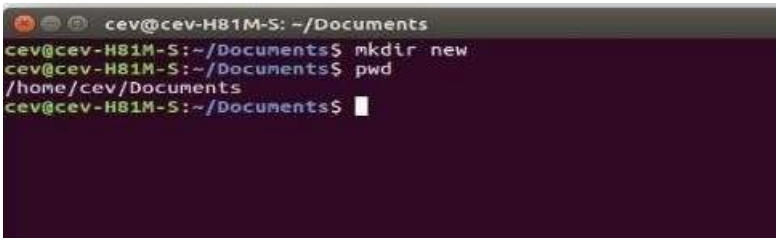
**7)cat**: It is used to list the contents of a file on the standard output.



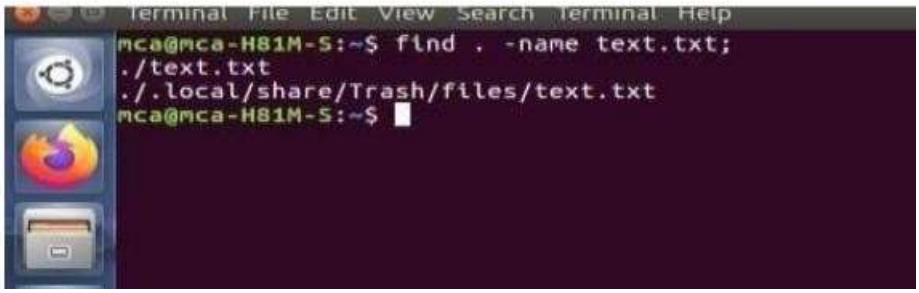

**8)cd**: It is used to navigate through the Linux files and directories.

**9)mkdir:** Create a new directory (folder).

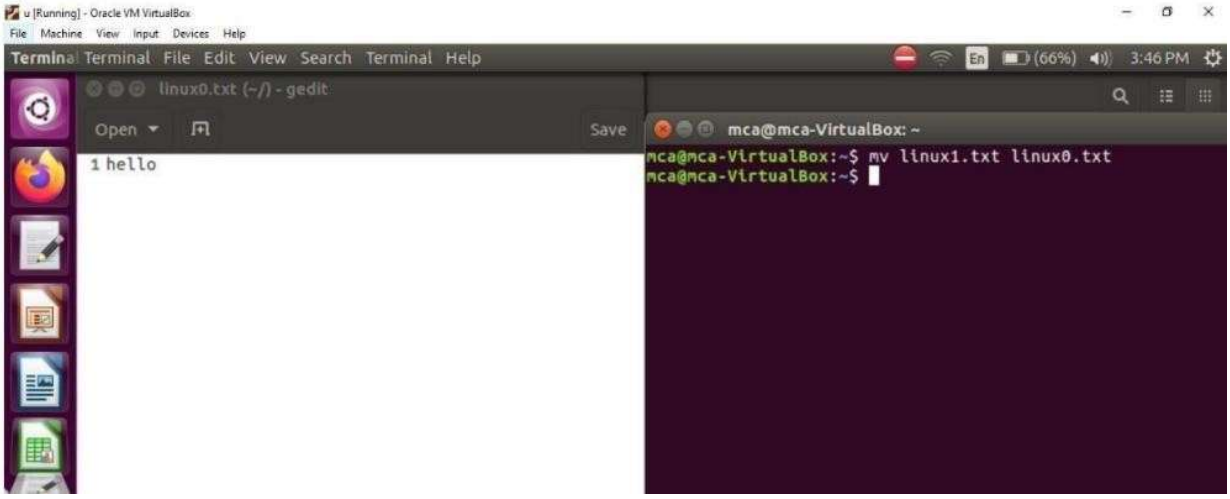
**10)pwd:** It print the current working directory path, starting from the root(/).




**11)find:** It is used to search and locate the list of files and directories based on conditions you specify for files that match the arguments.



**12)mv:** It is used to move one or more files or directories from one place to another in a file system like Unix.




**13)cp:** This command used to copy files or group of files or directory. For instance, the command cp scenery.jpg /home/username/Pictures would create a copy of scenery.jpg (from your current directory) into the Pictures directory. **cp –i :** will ask for user's consent in case of a potential file overwrite.

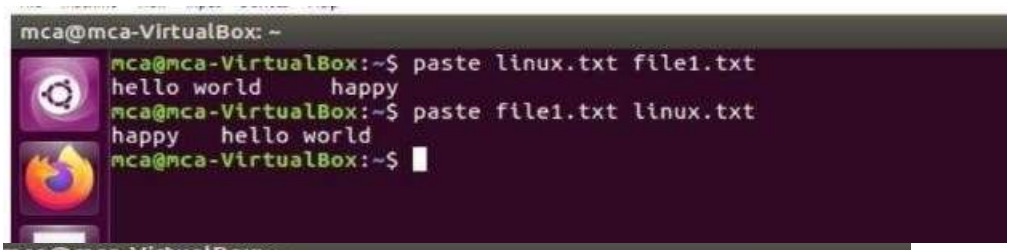
**14)rm:** It is used to remove objects such as files, directories, symbolic, links and so on from the file system.

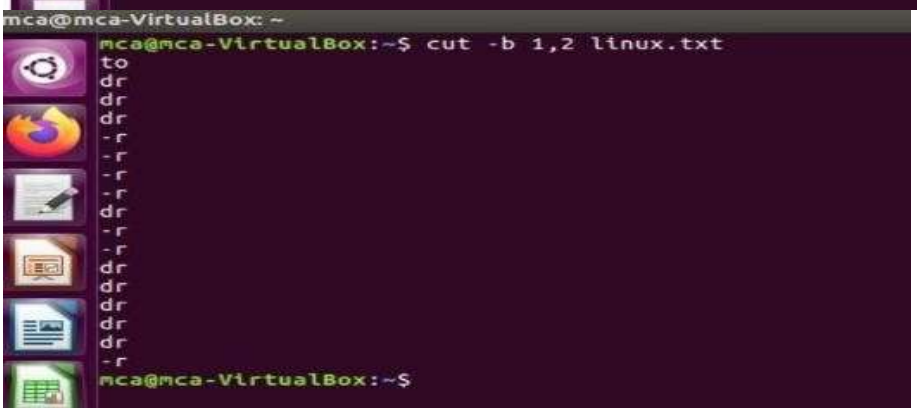
Eg: rm Music, will deletes the directory named Music.

**15)tar:** It is used for saving several files into an archive file.

**16)cut:** It is used for cutting out the sections from each line of files and writing the result to standard output.

**17)paste:** It is used to join horizontally by outputting lines consisting of lines from each file specified, separated by tab as delimiter, to standard output.




**18)head:** It present in all major Linux distributions which are used to print out data from the start of a file. For example, if you only want to show the first five lines, type head -n 5 filename. Ext.

**19)tail:** The basic functionality of Linux tail commands is to output the end of a file. For example, tail -n filename. Ext.

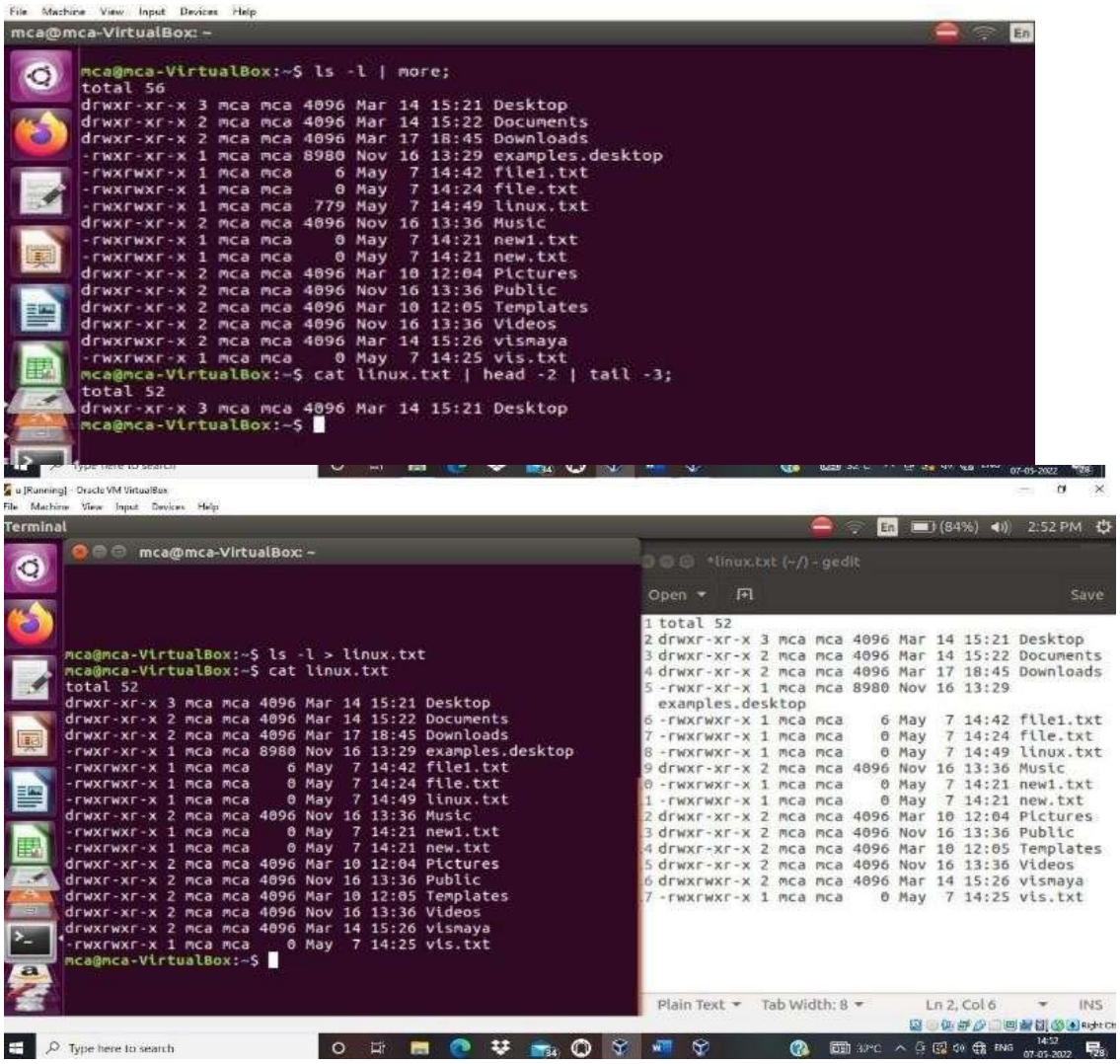
**20)grep**: Grep command is used to search through all the text in a given file.

**21)expr**: It was used to evaluate a given expression and display its corresponding output.

**22)chmod:** It is used to change the access permissions of files and directories. It stands for change mode. It cannot change the permission of symbolic links. Even, it ignores the symbolic links come across recursive directory traversal. Syntax: chmod <options> <permissions> <file name>

**23)chown:** It is used to change the files ownership, directory, or symbolic link for a user or group. In Linux, each file is associated with a corresponding owner or group. The Linux system may have multiple users. Every user has a unique name and user ID. If only a user is available in the system, the user will be the owner of each file. Syntax: chown [OPTION]… [OWNER] [: [GROUP]] FILE…

**24)redirection and piping:** Pipe is used to combine two or more commends and in this the output of one command and act as input to another command, and this command output may cut as input to the next command. Redirection in Linux command refers to the ability of the Linux operating system that allows as to change the standard input and standard output when executing a command on the terminal.



**25)useradd:** It is used to for adding /creating user accounts in Linux and other UNIX like operating systems.

```
root@cev-Veriton-M200-H81: /home/cev
cev@cev-Veriton-M200-H81:~$ su
Password:
root@cev-Veriton-M200-H81:/home/cev# useradd krish
root@cev-Veriton-M200-H81:/home/cev# passwd krish
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root@cev-Veriton-M200-H81:/home/cev#
```

**26)usermod:** It is used to modify existing user details , such username, password, home directory location, default shell, and more.

**27)userdel:** It is used to delete a user account and related files.

```
root@cev-Veriton-M200-H81: /home/cev
cev@cev-Veriton-M200-H81:~$ su
Password:
root@cev-Veriton-M200-H81:/home/cev# useradd krish
root@cev-Veriton-M200-H81:/home/cev# passwd krish
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root@cev-Veriton-M200-H81:/home/cev# whoami
root
root@cev-Veriton-M200-H81:/home/cev# su krish
krish@cev-Veriton-M200-H81:/home/cev$ whoami
krish
krish@cev-Veriton-M200-H81:/home/cev$ su
Password:
root@cev-Veriton-M200-H81:/home/cev# userdel -f krish
userdel: user krish is currently used by process 13777
root@cev-Veriton-M200-H81:/home/cev#
```

**28)passwd:** Passwd command used to change password for user accounts. The root user reserves the privilege to change the password for any user on the system, while a normal user can only change the account password for his or her own account. Syntax: passwd [options] [username] Eg: Command: passwd

**29)df:** It is used to display the disk space used in the file system.

```
mca@mca-H81M-S: ~
mca@mca-H81M-S:~$ df;
Filesystem     1K-blocks    Used Available Use% Mounted on
udev              914052       0    914052   0% /dev
tmpfs             187220    5940    181280   4% /run
/dev/sda2      144514712 9342692 127807976   7% /
tmpfs             936096     320    935776   1% /dev/shm
tmpfs               5120       4      5116   1% /run/lock
tmpfs             936096       0    936096   0% /sys/fs/cgroup
tmpfs             187220      64    187156   1% /run/user/1000
mca@mca-H81M-S:~$
```

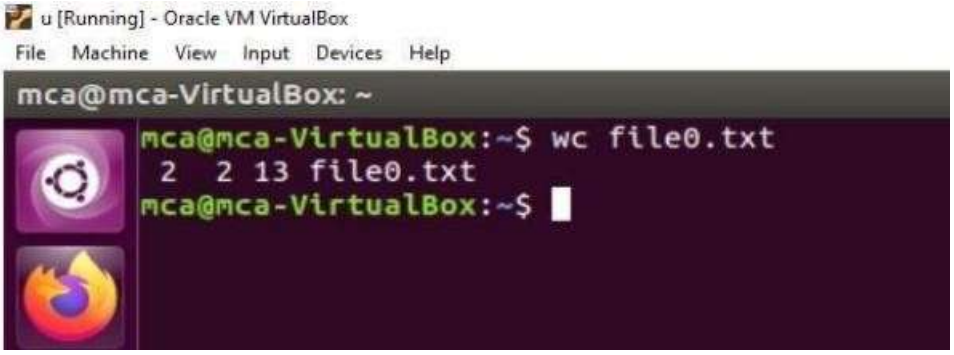
**30)top:** It shows the real-time view of running process in Linux and displays and kernel managed tasks. It's very useful to monitor system resource usage, especially knowing which process needs to be terminated because it consumes too many resources.

```
mca@mca-H81M-S: ~
mca@mca-H81M-S:~$ top;

top - 10:19:42 up  1:08,  1 user,  load average: 0.36, 0.13, 0.04
Tasks: 179 total,   1 running, 178 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.5 us,  0.2 sy,  0.0 ni, 98.8 id,  0.5 wa,  0.0 hi,  0.0 si,  0.0 st
KiB Mem :  1872192 total,   182572 free,   719980 used,   969640 buff/cache
KiB Swap:  3905532 total,  3905532 free,        0 used.   650488 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU %MEM     TIME+ COMMAND
  814 root      20   0  387012  56188  44448 S   0.7  3.0   0:30.69 Xorg
 1705 mca       20   0 1497404 137576  63268 S   0.3  7.3   0:42.96 compiz
 4293 mca       20   0  664752  38392  27928 S   0.3  2.1   0:00.54 gnome-terminal-
 4426 mca       20   0   41804   3800   3160 R   0.3  0.2   0:00.04 top
    1 root      20   0  119880   6032   4088 S   0.0  0.3   0:00.94 systemd
    2 root      20   0       0      0      0 S   0.0  0.0   0:00.00 kthreadd
    3 root      20   0       0      0      0 S   0.0  0.0   0:00.02 ksoftirqd/0
    5 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 kworker/0:0H
    7 root      20   0       0      0      0 S   0.0  0.0   0:01.74 rcu_sched
    8 root      20   0       0      0      0 S   0.0  0.0   0:00.00 rcu_bh
    9 root      rt   0       0      0      0 S   0.0  0.0   0:00.00 migration/0
   10 root      rt   0       0      0      0 S   0.0  0.0   0:00.01 watchdog/0
   11 root      rt   0       0      0      0 S   0.0  0.0   0:00.01 watchdog/1
   12 root      rt   0       0      0      0 S   0.0  0.0   0:00.00 migration/1
   13 root      20   0       0      0      0 S   0.0  0.0   0:00.04 ksoftirqd/1
   15 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 kworker/1:0H
   16 root      20   0       0      0      0 S   0.0  0.0   0:00.00 kdevtmpfs
   17 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 netns
   18 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 perf
   19 root      20   0       0      0      0 S   0.0  0.0   0:00.00 khungtaskd
   20 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 writeback
   21 root      25   5       0      0      0 S   0.0  0.0   0:00.00 ksmd
   22 root      39  19       0      0      0 S   0.0  0.0   0:00.00 khugepaged
   23 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 crypto
   24 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 kintegrityd
   25 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 bioset
   26 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 kblockd
   27 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 ata_sff
   28 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 md
   29 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 devfreq_wq
   34 root      20   0       0      0      0 S   0.0  0.0   0:00.10 kswapd0
   35 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 vmstat
   36 root      20   0       0      0      0 S   0.0  0.0   0:00.00 fsnotify_mark
   37 root      20   0       0      0      0 S   0.0  0.0   0:00.00 ecryptfs-kthrea
```

**31)ps:** It is used to list the currently running processes and their PIDs along with some other information depends on different option.

```
mca@mca-H81M-S: ~
mca@mca-H81M-S:~$ ps;
  PID TTY          TIME CMD
 4335 pts/17   00:00:00 bash
 4346 pts/17   00:00:00 ps
mca@mca-H81M-S:~$
```

**32)wc:** word count. It is mainly used for counting purpose.




**33)ssh:** It instructs the system to establish an encrypted secure connection with the host machine.

To check the system containing ssh using the command;

**$ "ssh"**

The installation command on ssh is:

**$ "sudo apt-get install open ssh-server"** To check the system IP address using the command:

**$ "ipconfig"**

Ping command using to check working:

**$ "ping second system IP"**

To login second system using the given command:

**$ "ssh second system user@second system IP**
**$ "cd Desktop"**
**$ "ls"**

**34)scp:** It is used to copy files between servers in a secure way.
Command:
**$ "scp 2nd system file path 1st system user@1st system**
**IP:** 2nd system path to logout the connection using:
**$ "logout/cntrl+D"**
**35)ssh-keygen:** It is used to generate, manage, and convert authentication keys for "ssh".

**36)ssh-copy-id:** It uses the "ssh" protocol to connect to the target host and upload the "ssh" user key.

**RESULT**

Linux text editors can be used for editing text files, writing codes, updating user instruction files, and more. A text editor plays an important role while coding. So, it is important to select the best text editor. A text editor should not only be simple but also functional and should be good to work with. A Linux command is a program or utility that runs on the command line. A command line is an interface that accepts lines of text and processes them into instructions for your computer.

# EXPERIMENT 4

**Aim:** File system hierarchy is a common Linux distribution, file and device permissions, study of the system configuration files in /etc., familiarizing log files for system events, user activity, network events.

## FILE SYSTEM HIERARCHY

The Linux File Hierarchy Structure or the Filesystem Hierarchy Standard (FHS) defines the directory structure and directory contents in Unix-like operating systems. It is maintained by the Linux Foundation.

## LOG FILES AND CONFIGURATION FILES

Linux uses a set of configuration files, directories, programs, commands and daemons to create, store and recycle these log messages. Knowing where the system keeps its log files and how to make use of related commands can therefore help save valuable time during troubleshooting.

### USER TYPES

- **Root User:** Access to all the file on the system.
- **Regular User**: This is the normal user account.
- **Service User:** Using service to run process and execute functions.

### DIRECTORY STRUCTURE

**(Root):** Primary hierarchy root and root directory of the entire file system hierarchy.

- Every single file and directory start from the root directory
- The only root user has the right to write under this directory
- /root is the root user's home directory, which is not the same as /

**/bin:** Essential command binaries that need to be available in single-user mode; for all users, e.g., cat, ls, cp.

- Contains binary executables
- Common Linux commands you need to use in single-user modes are located under this directory.
- Commands used by all the users of the system are located here e.g., ps, ls, ping, grep, cp

**/boot :** Boot loader files, e.g., kernels, initrd.

- Kernel initrd, vmlinux, grub files are located under /boot
- Example: initrd.img-2.6.32-24-generic, vmlinuz-2.6.32-24-generic

**/dev :** Essential device files, e.g., /dev/null.

- These include terminal devices, usb, or any device attached to the system.
- Example: /dev/tty1, /dev/usbmon0

**/etc :** Host-specific system-wide configuration files.

- Contains configuration files required by all programs.
- This also contains startup and shutdown shell scripts used to start/stop individual programs.
- Example: /etc/resolv.conf, /etc/logrotate.conf.

**/home :** Users' home directories, containing saved files, personal settings, etc.

- Home directories for all users to store their personal files.
- example: /home/kishlay, /home/kv

**/lib :** Libraries essential for the binaries in /bin/ and /sbin/.

- Library filenames are either ld* or lib*.so. *
- Example: ld-2.11.1.so, libncurses.so.5.7

**/media** : Mount points for removable media such as CD-ROMs (appeared in FHS-2.3).

- Temporary mount directory for removable devices.
- Examples, /media/cdrom for CD-ROM; /media/floppy for floppy drives; /media/cdrecorder for CD writer

**/mnt :** Temporarily mounted filesystems.

- Temporary mount directory where sysadmins can mount filesystems.

**/opt :** Optional application software packages.

- Contains add-on applications from individual vendors.
- Add-on applications should be installed under either /opt/ or /opt/ sub-directory.

**/sbin :** Essential system binaries, e.g., fsck, init, route.

- Just like /bin, /sbin also contains binary executables.
- The Linux commands located under this directory are used typically by system administrator,for system maintenance purpose.
- Example: iptables, reboot, fdisk, ifconfig, swapon

- **/srv :** Site-specific data served by this system, such as data and scripts for web servers, data offered by FTP servers, and repositories for version control systems.
  - srv stands for service.
  - Contains server specific services related data.
  - Example, /srv/cvs contains CVS related data.

**/tmp :** Temporary files. Often not preserved between system reboots, and may be severely size restricted.

- Directory that contains temporary files created by system and users.
- Files under this directory are deleted when system is rebooted.

**/usr :** Secondary hierarchy for read-only user data; contains the majority of (multi)user utilities and applications.

- Contains binaries, libraries, documentation, and source-code for second level programs.
- /usr/bin contains binary files for user programs. If you can't find a user binary under /bin, look under /usr/bin. For example: at, awk, cc, less, scp

**/proc :** Virtual filesystem providing process and kernel information as files. In Linux, corresponds to a procfs mount. Generally, automatically generated and populated by the system, on the fly. • Contains information about system process.

- This is a pseudo filesystem contains information about running process. For example:/proc/{pid} directory contains information about the process with that particular pid.
- This is a virtual filesystem with text information about system resources. For example:/proc/uptime

## FILE PERMISSIONS

**READ:** The read permission refers to a user's capability to read the contents of the file.

**WRITE:** The write permission refers to a user's capability to write or modify a file or directory.

**EXECUTE:** The execute permission affects a user's capability to execute a file or view the contents of a directory.
Each letter denotes a particular permission:

- **r**: Read permission
- **w**: Write permission
- **x**: Execute permission ● **——:** No permission set

**RESULT**

Successfully completed Linux file permissions. Linux divides the file permissions into read, write and execute denoted by r, w, and x. This document explains how directory and file permissions on a UNIX or Linux machine are set and can be changed by the user.

# EXPERIMENT 5

**Aim**: Shell scripting: study bash syntax, environment variables, control constructs such as if, for and while, aliases and functions, accessing command line arguments passed to shell scripts. Study of startup scripts, login and logout scripts, familiarity with system and system 5 init scripts is expected.

## SHELL SCRIPTING

A shell script is a computer program designed to be run by the Unix/Linux shell which could be one of the following:

- The Bourne Shell
- The C Shell
- The Korn Shell
- The GNU Bourne-Again Shell

A shell is a command-line interpreter and typical operations performed by shell scripts include file manipulation, program execution, and printing text. In Linux, shells like bash and Korn support programming construct which are saved as scripts. These scripts become shell commands and hence many Linux commands are support.

- The "$" sign stands for a shell variable and echo will return the text whatever you typed in.
- "**#!**" is called she-bang and written at top of the script.

## PROGRAM 1

**AIM:** write a shell scripting program to display a given message?

**SOURCE CODE:**

```
#!/bin/bash
read msg
echo $msg
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg1.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg1.sh
hello world!
hello world!
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

## PROGRAM 2

**AIM:** write a shell script to print whether two numbers are equal or not?

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter first number:" n1
read -p "enter second number:" n2
if [ "$n1" -eq "$n2" ];
then
echo "two numbers are equal:"
else
echo "two numbers are not equal"
fi
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg2.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg2.sh
enter first number:5
enter second number:6
Two numbers are not equal
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

## PROGRAM 3

**AIM:** create a shell script to perform integer arithmetic operation?

**SOURCE CODE:**

```
x=0
i="y"
while [ "$i" == "y" ];
do
read -p "enter first number:" a
read -p "enter second number:" b
echo "1. addition"
echo "2. subtraction"
echo "3. multiplication"
echo "4. division"
read -p "enter your choice:" c
case $c in
```

```
1)
  x=$((a+b))
  echo "sum:" $x
  ;;
2)
  x=$((a-b))
  echo "Difference:" $x
  ;;
3)
  x=$((a*b))
  echo "multiplication:" $x
  ;;
4)
  x=$(echo "scale=2; $a/$b" | bc)
  echo "Division:" $x
  ;;
*)
  echo "invalid option"
  ;;
esac
echo "Do you want to continue: [y/n]"
read i
if [ "$i" != "y" ];
then
exit 0
fi
done
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg3.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg3.sh
enter first number:5
enter second number:6
Addition=11
Substraction=-1
Multiplocation=30
Modulo division=5
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

## PROGRAM 4

**AIM:** write a shell script to getting input details Like name, roll no, and marks and print them?

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter your name:" name
read -p "enter your roll no:" rollno
read -p "enter your mark:" mark
echo "name:" $name
echo "roll no:" $rollno
echo "mark:" $mark
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg4.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg4.sh
enter your name:anu
enter your roll no:24
enter your mark:95
name= anu
roll no= 24
mark= 95
```

## PROGRAM 5

**AIM:** write a shell program to swap two numbers?

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter first number:" a
```

```
read -p "enter second number:" b
echo "before swap first number:" $a
echo "before swap second number:" $b
temp=$a
a=$b
b=$temp
echo "after  the swap first number=" $a
echo "after  the swap second number=" $b
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg5.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg5.sh
enter first number:5
enter second number:9
before swap first number=5
before swap second number=9
after swap first number= 9 echo after swap second number= 5
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

## PROGRAM 6

**AIM: :** Write a shell program to check whether the input number is prime or not?

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter the number:" a
i=2
f=0
while [ $i -lt $a ];
do
s=$(expr $a % $i)
if [ $s -eq $f ];
then
echo "not prime"
exit
else
i=$(expr $i + 1)
fi
done
echo "prime number"
```

**OUTPUT:**

```
student@cev:~/alfa ads$ touch prg6.sh
student@cev:~/alfa ads$ nano prg6.sh
student@cev:~/alfa ads$ bash prg6.sh
enter the number:
7
prime number
student@cev:~/alfa ads$
```

## PROGRAM 7

**AIM:** write a shell program to find the square and cube of a number?

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter the number:" a
square=$(($a * $a))
cube=$(($a * $a *$a ))
echo "square="$square
echo "cube="$cube
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg7.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg7.sh
enter the number:5
square=25
cube=125
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

## PROGRAM 8

**AIM:** write a shell program switch whether the given number is odd or even?

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter a number" a
b=$(( $a % 2 ))
if [ $b -eq 0 ]
then
echo "then number is even"
else
echo "the number is odd"
fi
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg8.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg8.sh
enter a number: 5
the number is odd
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

**PROGRAM 9**

**AIM:** Write a shell program to find the minimum among four values?

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter the first number:" a
read -p "enter the second number:" b
read -p "enter the third number:" c
read -p "enter the fourth number:" d
if [ $a -lt $b ] && [ $a -lt $c ] && [ $a -lt $d ]
then
echo "$a is smallest number"
elif [ $b -lt $a ] && [ $b -lt $c ] && [ $b -lt $d ]
then
echo "$b is smallest number"
elif [ $c -lt $b ] && [ $c -lt $a ] && [ $c -lt $d ]
then
```

```
echo "$c is smallest number"
else
echo "$d is smallest number"
fi
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg9.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg9.sh
enter first number:5
enter second number:7
enter third number:10
enter fourth number:15
5 is smallest number:
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

## PROGRAM 10

**AIM:** write a shell script to print sum of N natural numbers?

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter a number:" n
sum=0
i=1
while [ $i -lt $n ]
do
sum=$(( sum + i ))
i=$((i + 1))
done
echo "result =" $sum
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg10.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg10.sh
Enter a number
5
Result = 15
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

## PROGRAM 11

**AIM:** Write a shell program to find area of a triangle?

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter the breadth:" b
read -p "enter the height:" h
area=$(((($b * $h) / 2))
echo "area of the triangle : $area"
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg11.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg11.sh
enter the height of the triangle:5
enter the bredth of the triangle:6
area of the triangle is: 15
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

## PROGRAM 12

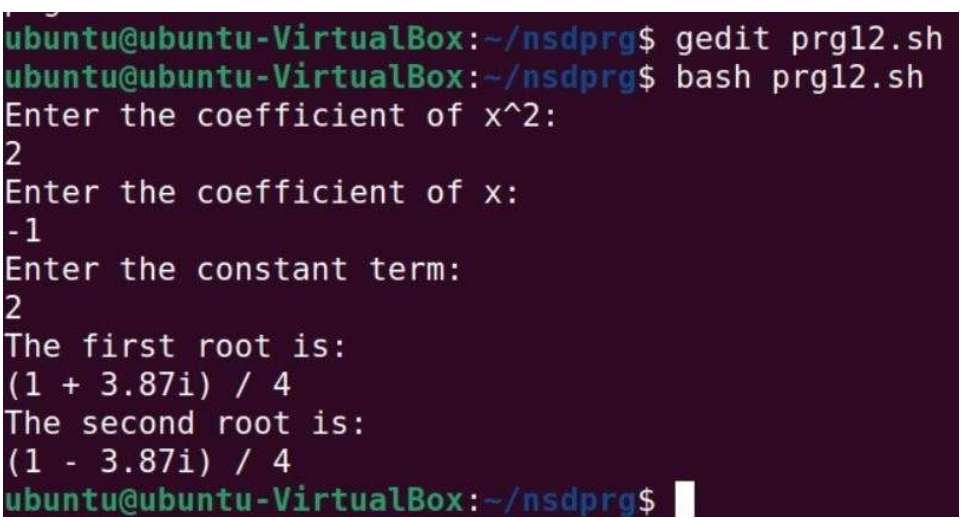
**AIM:** Write a shell program to find root of the quadratic equation?

**SOURCE CODE:**

```
echo Enter the coefficient of x^2:
read a
echo Enter the coefficient of x:
read b
echo enter the constant term:
read c
f=`echo "-($b)" |bc`
p=`expr 2 \*  $a`
if [ $a -ne 0 ]
  then
  d=$(echo "$b*$b - 4*$a*$c" | bc)
  if [ $d -lt 0 ]
      then
      x=`echo "-($d)" | bc -l`
```

```
        s=`echo "scale=2; sqrt ($x)" | bc -l`
        echo the first root is:
        echo "($f + $s i)/$p"
        echo the second root is:
        echo "($f - $s i)/$p"
    elif [ $d -eq 0 ]
    then
       res=`expr $f / $p`
       echo the root is:$res
    else
       s=`echo "scale=2; sqrt($d)" | bc`
       res1=`echo "scale=2; ($f + $s)/($p)"|bc -l`

       res2=`echo "scale=2;($f-$s)/($p)"|bc -l`
       echo the first root is:$res1
       echo the second root is:$res2
    fi
    else
    echo coefficient of x^2 can not be 0.
fi
```

**OUTPUT:**



## PROGRAM 13

**AIM:** program to find the factorial of a given number?

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter the number" n
fact=1
while [ $n -gt 1 ]
do
fact=$(( $fact * $n ))
n=$(( n - 1 ))
done
echo "the factorial =" $fact
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg13.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg13.sh
enter a number:5
factorial=120
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

## PROGRAM 14

**AIM:** Write a shell program to reverse a number?

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter a three digit number:" n
while [ $n -ne 0 ]
do
rev=$rev$(( n % 10 ))
n=$(( n / 10 ))
done
echo "reverse of the number: " $rev
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg14.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg14.sh
enter a three digit number:370
reverse of the number is: 073
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

## PROGRAM 15

**AIM:** program to print Fibonacci series?

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter the limit : " n
x=0
y=1
echo $x
echo $y
i=2
while [ $i -lt $n ]
do
sum=$(( $x + $y ))
echo $sum
x=$y
y=$sum
i=$(( $i + 1 ))
done
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg15.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg15.sh
enter the limit: 10
0
1
1
2
3
5
8
13
21
34
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

## PROGRAM 16

**AIM**: script program to get whether the given year is leap year or not?

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter the year : " n
if (( $n % 4 == 0 && $n % 100 != 0 || $n % 400 == 0 ))
then
echo "$n is leap year"
else
echo "$n is not a leap year"
fi
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg16.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg16.sh
enter the year :
2021
2021 is not leap year
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg16.sh
enter the year :
2024
2024 is leap year
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

## PROGRAM 17

**AIM :** Write a shell script to count a number of words and line in a file?

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter the file name" n
if [ -f "$n" ];then
line_c=0
word_c=0
while read -r l ;do
line_c=$(( line_c + 1 ))
for i in $l;do
word_c=$(( word_c + 1 ))
done
done < "$n"
echo "number of line $line_c"
echo "number of word $word_c"
else
echo "file not found"
fi
```

**OUTPUT:**

## PROGRAM 18

**AIM:** Write a shell script program to display message “good morning” , ”good afternoon” , “good evening “ and “good night” according to system time?

**SOURCE CODE:**

```
#!/bin/bash
h=$(date +"%H")
if [ $h -gt 6 -a $h -le 12 ]
then
echo good morning
elif [ $h -gt 12 -a $h -le 16 ]
then
echo good afternoon
elif [ $h -gt 16 -a $h -le 20 ]
then
echo good evening
else
echo good night
fi
```

**OUTPUT:**

```
student@cev:~/varun/Pg18$ vim pg18
student@cev:~/varun/Pg18$ bash pg18
good morning
student@cev:~/varun/Pg18$
```

## PROGRAM 19

**AIM:** Shell script to compare two files. If the files are identical delete one

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter the first file: " file1
read -p "enter the second file: " file2
if cmp -s "$file1 " "$file2"; then
    echo "the files are identical.Deleting $file2..."
    rm "$file2"
```

```
else
   echo "the file are different."
Fi
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg19.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg19.sh

Enter the first file: /home/ubuntu/nsdprg/abc.txt
Enter the second file: /home/ubuntu/nsdprg/cba.txt
The files are identical. Deleting /home/ubuntu/nsdprg/cba.txt
...
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

## PROGRAM 20

**AIM:** Shell script to rename all ordinary files in a directory with filename.PID of shell

**SOURCE CODE:**

```
#!/bin/bash
pid=$$
for file in *
do
        if [ -f"$file" ];then
                mv "$file" "$file.$pid"
        echo "renamed $file to $file.$pid"
        fi
done
for f in *.5478; do mv "$f" "${f%.5478}"; done
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg/prg20$ ls
file1.txt  file2.txt  prg20.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg/prg20$ gedit prg20.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg/prg20$ bash prg20.sh
Renamed file1.txt to file1.txt.5344
Renamed file2.txt to file2.txt.5344
Renamed prg20.sh to prg20.sh.5344
ubuntu@ubuntu-VirtualBox:~/nsdprg/prg20$ ~
```

## PROGRAM 21

**AIM:** Shell script to display the content of a directory file in windows style

**SOURCE CODE:**

```
#!/bin/bash
read -p "enter the directory path: " dir
if [ ! -d "$dir" ]; then
  echo "Directory $dir does not exist."
  break
 fi
echo "Directory listing fo $dir:"
ls -l "$dir"
```

**OUTPUT:**

```
ubuntu@ubuntu-VirtualBox:~/nsdprg$ gedit prg21.sh
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg21.sh
Enter the directory path: /home/ubuntu/nsdprg/prg20
Directory listing for /home/ubuntu/nsdprg/prg20:
file1.txt.5344
file2.txt.5344
prg20.sh.5344
ubuntu@ubuntu-VirtualBox:~/nsdprg$ bash prg21.sh
Enter the directory path: /home/ubuntu/nsdprg/text
Directory /home/ubuntu/nsdprg/text does not exist.
ubuntu@ubuntu-VirtualBox:~/nsdprg$
```

# EXPERIMENT 6

**Aim:** Analyzing network packet stream using Wireshark. Perform basic network service test using nc.

## WIRESHARK

Wireshark is an open-source network protocol analyzer tool indispensable for system administration and security. It drills down and displays data travelling on the network. Wireshark allows you to either capture live network packets or to save it for offline analysis.

Some of the tasks one can perform with Wireshark are

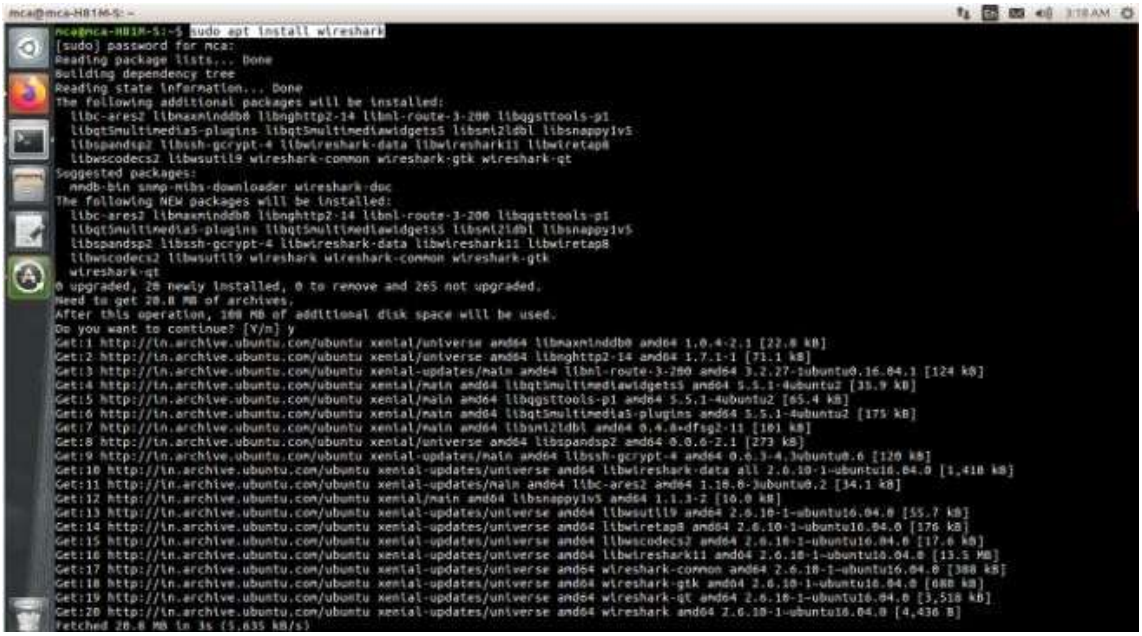
- Capturing and finding traffic passing through your network
- Inspection of hundreds of different protocols
- Live capture of traffic/offline analysis
- Troubleshooting dropped packets and latency problems
- Looking at attempts of attacks or malicious activities Installing Wireshark.

For installing Wireshark, you will need to add the "Universe" repository. Issue the following command in Terminal to do so:

❖ **sudo add-apt-repository universe**

Now issue the following command in Terminal to install Wireshark on your system:

❖ **sudo apt install Wireshark**




When prompted for a password, type sudo password.

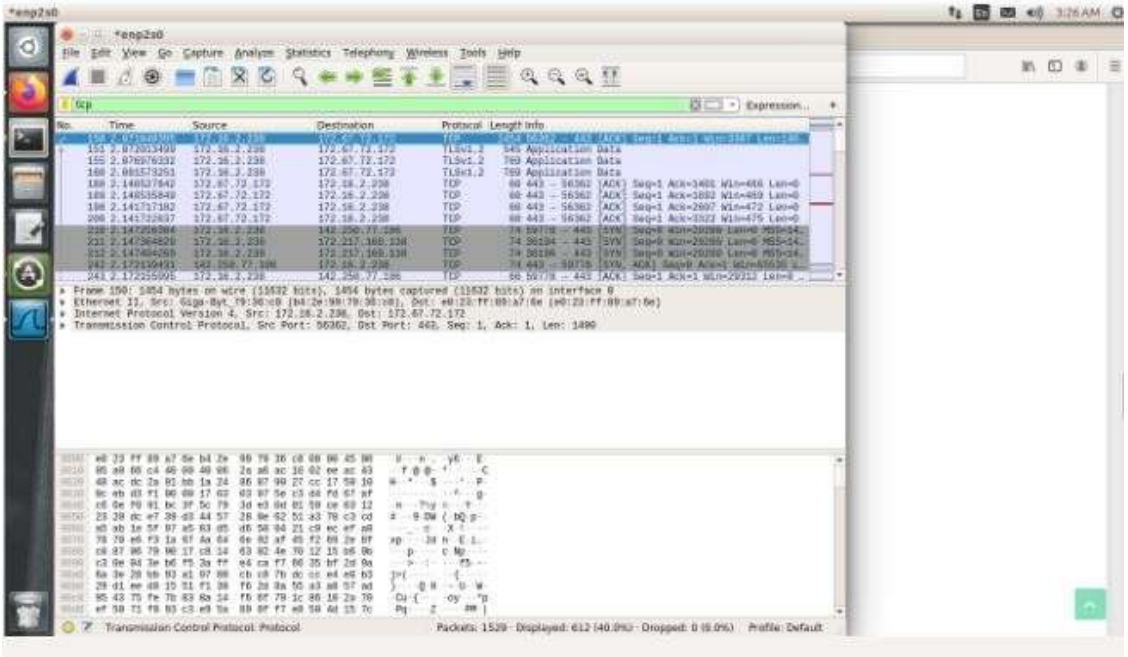
❖ **wireshark --version**

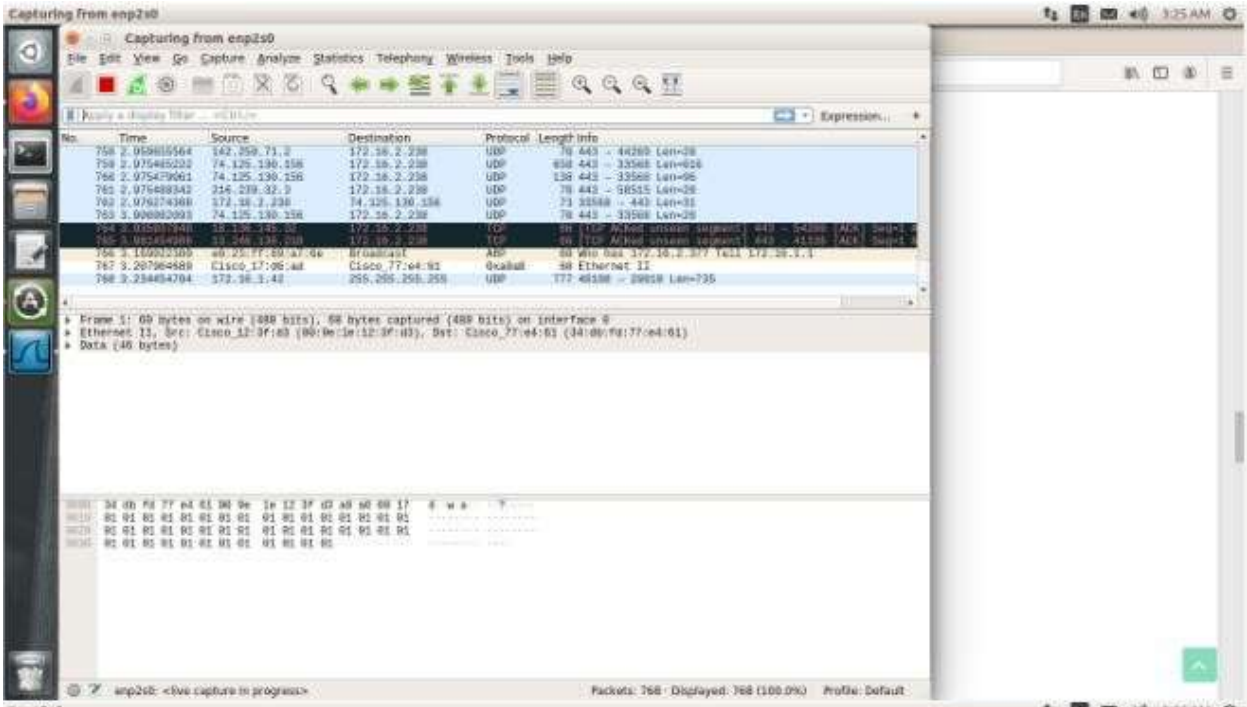
If Wireshark has installed successfully, you will a similar output displaying the version of Wireshark installed.

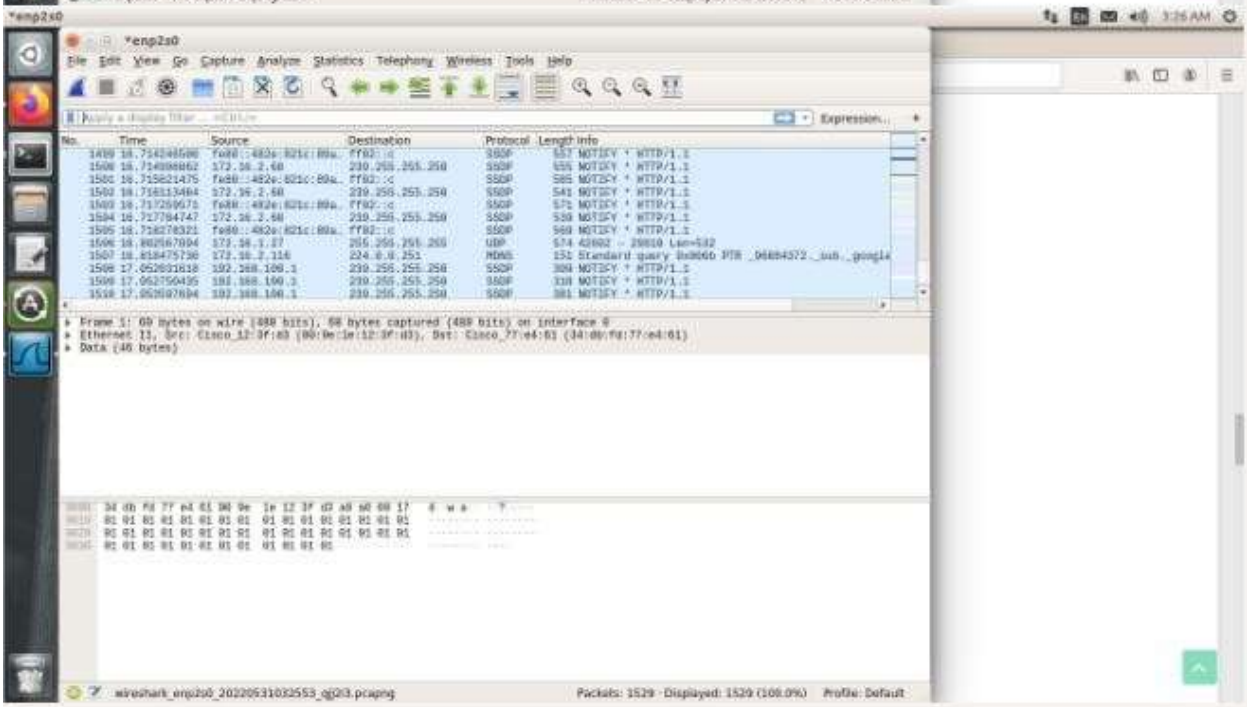
❖ **sudo wireshark**

If you are logged in as a root user, you can also launch Wireshark form the GUI. Hit the super key and type Wireshark in the search bar. When the icon for the Wireshark appears, click on it to launch it.

Wireshark is a powerful tool with plenty of features. Here we will just go through the basics of the two important features that are: packet capture and display filter.




Capturing from enp2s0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Expression...
No. Time Source Destination Protocol Length Info
Packets: 768 · Displayed: 768 (100.0%) Profile: Default


*enp2s0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Expression...
No. Time Source Destination Protocol Length Info
Packets: 1529 · Displayed: 1529 (100.0%) Profile: Default

# EXPERIMENT 7

**Aim**: Introduction to hypervisor and VMs, Xen or KVM, Introduction to containers: Docker, installation and deployment.

## DOCKER

Docker container most commonly used in software updation. Docker is an application that simplifies the process of managing application processes in containers. Containers let you run your applications in resource-isolated processes. They're similar to virtual machines, but containers are more portable, more resource-friendly, and more dependent on the host operating system.

Docker is a set of platform-as-a-service products used to launch and manage containers. Developers use Docker containers for developing and deploying applications because they provide isolated, lightweight, virtual environments.

## HYPERVISOR:

A hypervisor, also known as a virtual machine monitor or VMM, is software that creates and runs virtual machines (VMs). A hypervisor allows one host computer to support multiple guest VMs by virtually sharing its resources, such as memory and processing.

Hypervisors are two types,

Type 2 hypervisors are that Type 1 runs on bare metal and Type 2 runs on top of an operating system. Each hypervisor type also has its own pros and cons and specific use cases. Virtualization works by abstracting physical hardware and devices from the applications running on that hardware.

## VIRTUAL MACHINES

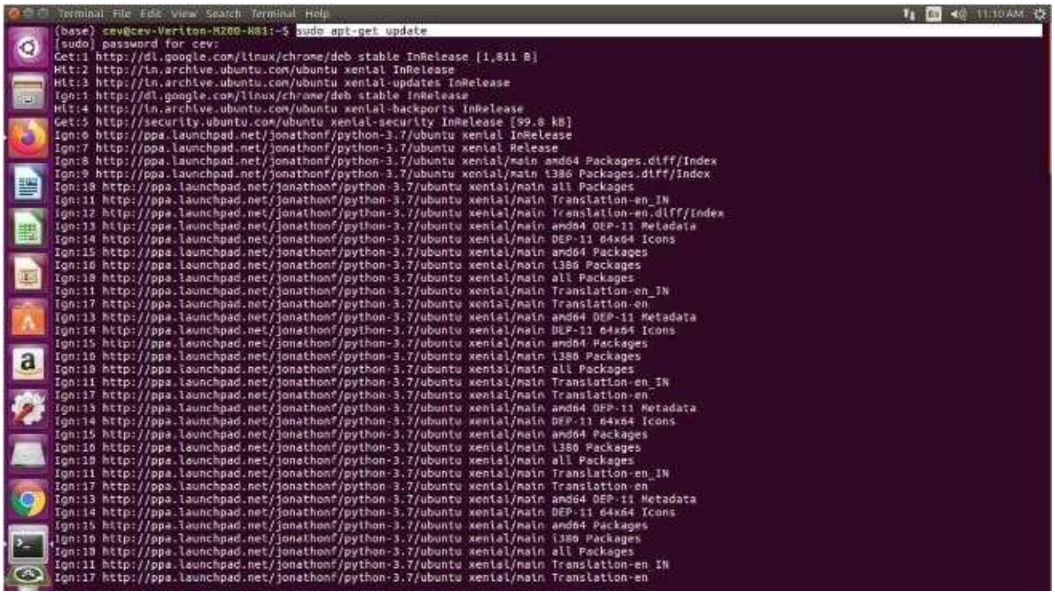
A virtual machine (VM) is a virtual environment that functions as a virtual computer system with its own CPU, memory, network interface, and storage, created on a physical hardware system.

## VIRTUALIZATION

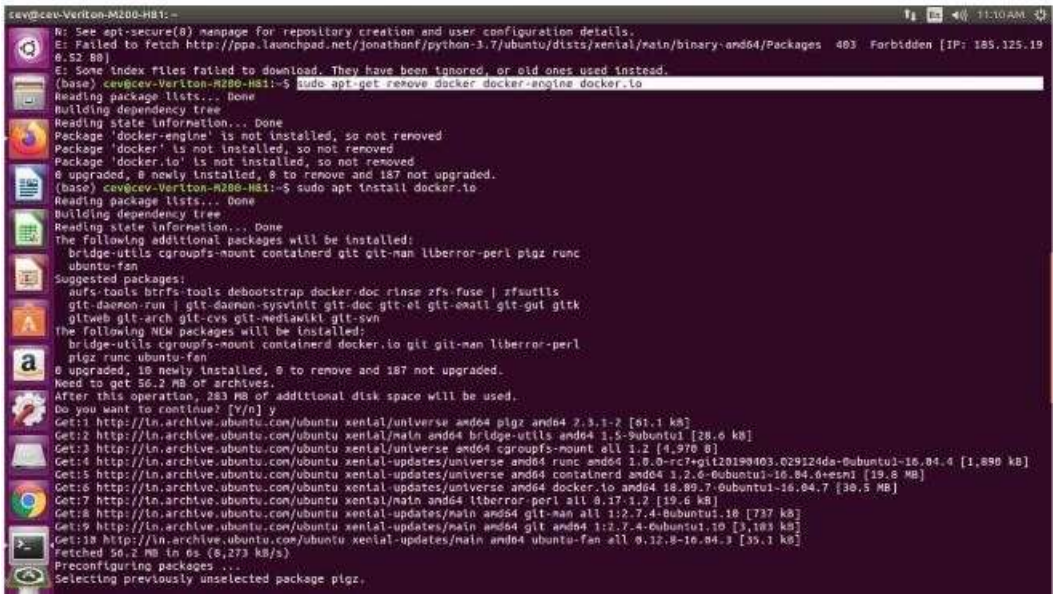
Network Virtualization (NV) refers to abstracting network resources that were traditionally delivered in hardware to software. NV can combine multiple physical networks to one virtual, software-based network, or it can divide one physical network into separate, independent virtual networks.

## INSTALLATION:

First, update your existing list of packages:

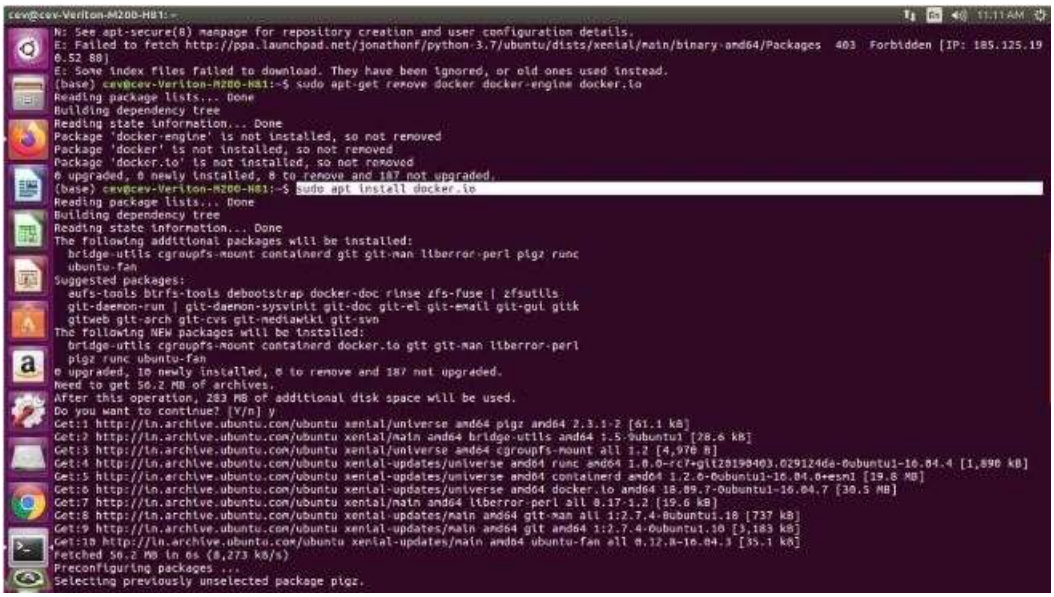



If is is already exist. Then remove Docker container.




Now let's install Docker on Ubuntu 20.04. Run the following command in the terminal window:

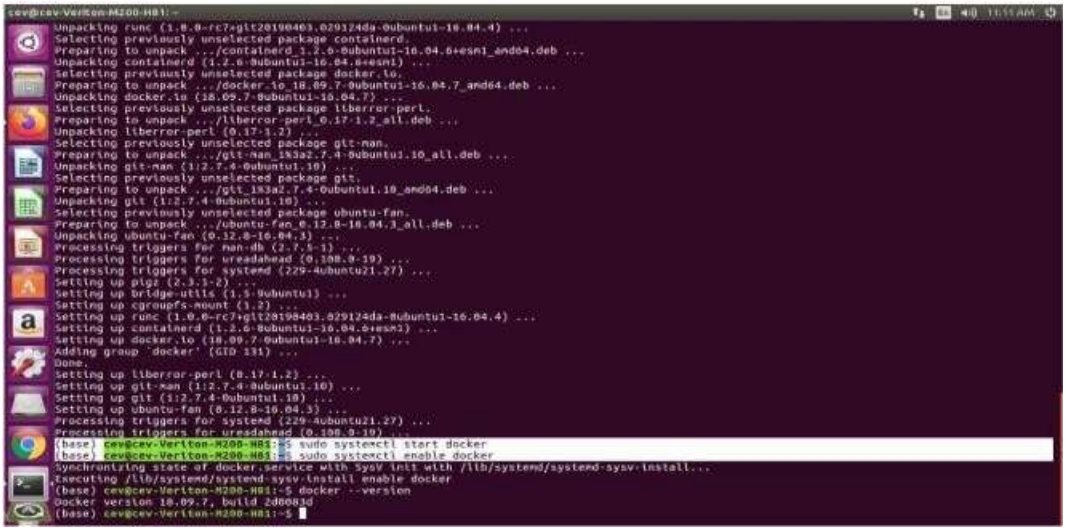
- **sudo apt install docker.io**

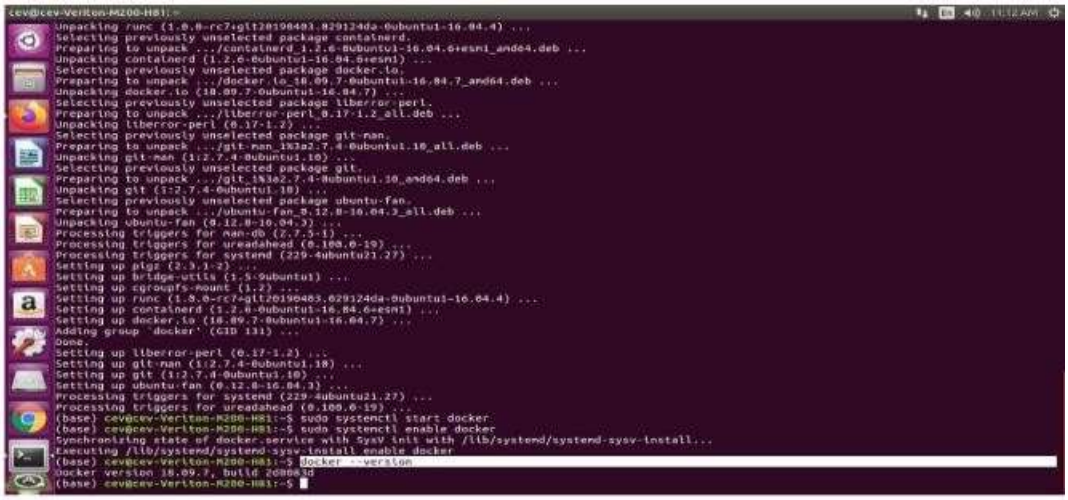



To verify the installation check the Docker version:

❖ **Docker –version**

- Start the Docker service by running:
- Then, enable it to run at startup:
- To check the status of the service, run:
- The output should verify Docker is **active (running)**.






**RESULT:**

Docker containers are isolated virtual environments that run based on the Docker image assigned to them. Docker container installed successfully.